

Q1: The fractional change in the magnetic field intensity at a distance 'r' from centre on the axis of current carrying coil of radius 'a' to the magnetic field intensity at the centre of the same coil is:

(Take $r < a$)

(A) $\frac{3}{2} \frac{a^2}{r^2}$

(B) $\frac{2}{3} \frac{a^2}{r^2}$

(C) $\frac{2}{3} \frac{r^2}{a^2}$

(D) $\frac{3}{2} \frac{r^2}{a^2}$

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Solution:

$$B_{axis} = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}$$

$$B_{centre} = \frac{\mu_0 i}{2R}$$

$$\therefore B_{centre} = \frac{\mu_0 i}{2a}$$

$$\therefore B_{axis} = \frac{\mu_0 i a^2}{2(a^2 + r^2)^{3/2}}$$

$\therefore$ fractional change in magnetic field =

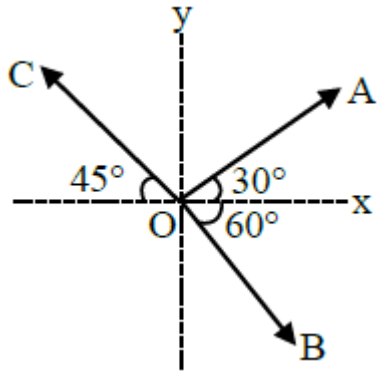
$$\frac{\frac{\mu_0 i}{2a} - \frac{\mu_0 i a^2}{2(a^2 + r^2)^{3/2}}}{\frac{\mu_0 i}{2a}} = 1 - \frac{1}{\left[1 + \left(\frac{r^2}{a^2}\right)\right]^{3/2}}$$
$$\approx 1 - \left[1 - \frac{3}{2} \frac{r^2}{a^2}\right] = \frac{3}{2} \frac{r^2}{a^2}$$

$$\text{Note : } \left(1 + \frac{r^2}{a^2}\right)^{-3/2} \approx \left(1 - \frac{3}{2} \frac{r^2}{a^2}\right)$$

[True only if $r \ll a$]

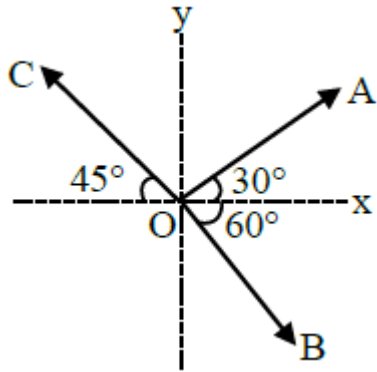
Hence option (d) is the most suitable option

Q2: The magnitude of vectors $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ in the given figure are equal. The direction of $\vec{OA} + \vec{OB} - \vec{OC}$ with x-axis will be :-



- (A) $\tan^{-1} \frac{(1-\sqrt{3}-\sqrt{2})}{(1+\sqrt{3}+\sqrt{2})}$
- (B) $\tan^{-1} \frac{(\sqrt{3}-1+\sqrt{2})}{(1+\sqrt{3}-\sqrt{2})}$
- (C) $\tan^{-1} \frac{(\sqrt{3}-1+\sqrt{2})}{(1-\sqrt{3}+\sqrt{2})}$
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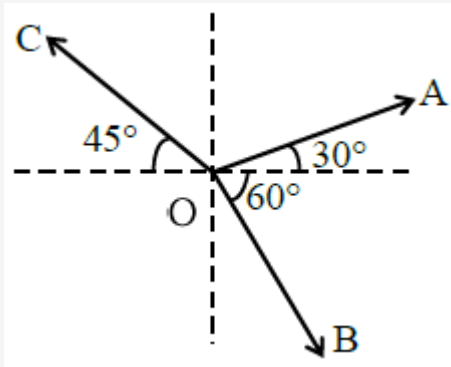
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(D) $\tan^{-1} \frac{(1+\sqrt{3}-\sqrt{2})}{(1-\sqrt{3}-\sqrt{2})}$

Solution:



Let magnitude be equal to λ .

$$\vec{OA} = \lambda [\cos 30^\circ \hat{i} + \sin 30^\circ \hat{j}] = \lambda \left[\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j} \right]$$

$$\vec{OB} = \lambda [\cos 60^\circ \hat{i} - \sin 60^\circ \hat{j}] = \lambda \left[\frac{1}{2} \hat{i} - \frac{\sqrt{3}}{2} \hat{j} \right]$$

$$\vec{OC} = \lambda [\cos 45^\circ (-\hat{i}) + \sin 45^\circ \hat{j}] = \lambda \left[-\frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{j} \right]$$

$$\begin{aligned} &\therefore \vec{OA} + \vec{OB} - \vec{OC} \\ &= \lambda \left[\left(\frac{\sqrt{3}}{2} + 1 \right) \hat{i} + \left(\frac{1}{2} - \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \right) \hat{j} \right] \end{aligned}$$

$\therefore$ Angle with x-axis

$$\begin{aligned}\tan^{-1} \left[\frac{\frac{1}{2} - \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}}{\frac{\sqrt{3}}{2} + \frac{1}{2} + \frac{1}{\sqrt{2}}} \right] &= \tan^{-1} \left[\frac{\sqrt{2} - \sqrt{6} - 2}{\sqrt{6} + \sqrt{2} + 2} \right] \\ &= \tan^{-1} \left[\frac{1 - \sqrt{3} - \sqrt{2}}{\sqrt{3} + 1 + \sqrt{2}} \right]\end{aligned}$$

Hence option (a)

Q3: Car B overtakes another car A at a relative speed of 40 ms^{-1} . How fast will the image of car B appear to move in the mirror of focal length 10 cm fitted in car A, when the car B is 1.9 m away from the car A?

- (A) 4 ms^{-1}
- (B) 0.2 ms^{-1}
- (C) 40 ms^{-1}
- (D) 0.1 ms^{-1}

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- (A) 4 ms^{-1}
- (B) 0.2 ms^{-1}
- (C) 40 ms^{-1}
- (D) 0.1 ms^{-1}

Solution:



Mirror used is convex mirror (rear-view mirror)

$$\therefore V_{I/m} = -m^2 V_{O/m}$$

Given,

$$V_{O/m} = 40 \text{ m/s}$$

$$m = \frac{f}{f - u} = \frac{10}{10 + 190} = \frac{10}{200}$$

$$\therefore V_{I/m} = -\frac{1}{400} \times 40 = -0.1 \text{ m/s}$$

$\therefore$ Car will appear to move with speed 0.1 m/s.

Hence option (d)

Q4: Inside a uniform spherical shell :

- (a) the gravitational field is zero
- (b) the gravitational potential is zero
- (c) the gravitational field is same everywhere
- (d) the gravitation potential is same everywhere
- (e) all of the above

Choose the most appropriate answer from the options given below:

- (A) (a), (c) and (d) only
- (B) (e) only
- (C) (a), (b) and (c) only
- (D) (b), (c) and (d) only

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Solution:

Inside a spherical shell, gravitational field is zero and hence potential remains same everywhere
Hence option (a)

Q5: Two narrow bores of diameter 5.0 mm and 8.0 mm are joined together to form a U-shaped tube open at both ends. If this U-tube contains water, what is the difference in the level of two limbs of the tube.

[Take surface tension of water $T = 7.3 \times 10^{-2} \text{ Nm}^{-1}$, angle of contact $= 0$, $g = 10 \text{ ms}^{-2}$ and density of water $= 1.0 \times 10^3 \text{ kg m}^{-3}$]

- (A) 3.62 mm
- (B) 2.19 mm
- (C) 5.34 mm
- (D) 4.97 mm

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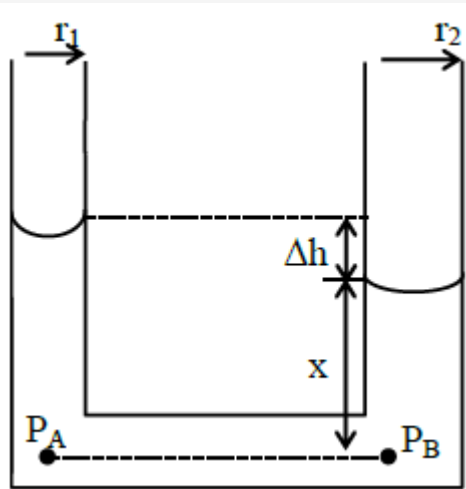
(A) 3.62 mm

(B) 2.19 mm

(C) 5.34 mm

(D) 4.97 mm

Solution:



We have $P_A = P_B$. [Points A & B at same horizontal level]

$$\therefore P_{atm} - \frac{2T}{r_1} + \rho g(x + \Delta h) = P_{atm} - \frac{2T}{r_2} + \rho g x$$

$$\therefore \rho g \Delta h = 2T \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$= 2 \times 7.3 \times 10^{-2} \left[\frac{1}{2.5 \times 10^{-3}} - \frac{1}{4 \times 10^{-3}} \right]$$

$$\therefore \Delta h = \frac{2 \times 7.3 \times 10^{-2} \times 10^3}{10^3 \times 10} \left[\frac{1}{2.5} - \frac{1}{4} \right]$$

$$= 2.19 \times 10^{-3} m = 2.19 mm$$

Hence option (b)

Q6: An electric appliance supplies 6000 J/min heat to the system. If the system delivers a power of 90W. How long it would take to increase the internal energy by $2.5 \times 10^3 J$?

- (A) $2.5 \times 10^2 s$
- (B) $4.1 \times 10^1 s$
- (C) $2.4 \times 10^3 s$
- (D) $2.5 \times 10^1 s$

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(C) $2.4 \times 10^3 s$

(D) $2.5 \times 10^1 s$

Solution:

$$\Delta Q = \Delta U + \Delta W$$

$$\frac{\Delta Q}{\Delta t} = \frac{\Delta U}{\Delta t} + \frac{\Delta W}{\Delta t}$$

$$\frac{6000}{60} \frac{J}{\text{sec}} = \frac{2.5 \times 10^3}{\Delta t} + 90$$

$$\Delta t = 250 \text{ sec}$$

Option (a)

Q7: An inductor coil stores 64 J of magnetic field energy and dissipates energy at the rate of 640 W when a current of 8A is passed through it. If this coil is joined across an ideal battery, find the time constant of the circuit in seconds :

- (A) 0.4
- (B) 0.8
- (C) 0.125
- (D) 0.2

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- (A) 0.4
- (B) 0.8
- (C) 0.125
- (D) 0.2

Solution:

$$U = \frac{1}{2}Li^2 = 64 \Rightarrow L = 2$$

$$i^2 R = 640$$

$$R = \frac{640}{(8)^2} = 10$$

$$\tau = \frac{L}{R} = \frac{1}{5} = 0.2$$

Option (d)

Q8: A series LCR circuit driven by 300 V at a frequency of 50 Hz contains a resistance $R = 3 \text{ k}\Omega$, an inductor of inductive reactance $X_L = 250 \pi \Omega$ and an unknown capacitor. The value of capacitance to maximize the average power should be : (Take $\pi^2 = 10$)

- (A) $4 \mu F$
- (B) $25 \mu F$
- (C) $400 \mu F$
- (D) $40 \mu F$

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Solution:

For maximum average power

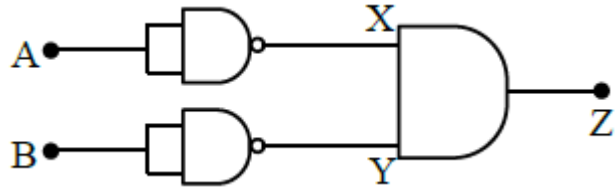
$$X_L = X_C$$

$$250\pi = \frac{1}{2\pi(50)C}$$

$$C = 4 \times 10^{-6} F$$

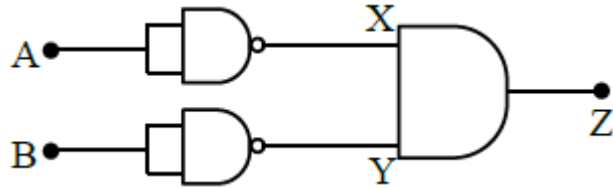
Option (a)

Q9: Identify the logic operation carried out by the given circuit :-



- (A) OR
- (B) AND
- (C) NOR
- (D) NAND

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Solution:

A	B	X	Y	Z
1	1	0	0	0
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

Option (c)

Q10: A particular hydrogen like ion emits radiation of frequency 2.92×10^{15} Hz when it makes transition from $n = 3$ to $n = 1$. The frequency in Hz of radiation emitted in transition from $n = 2$ to $n = 1$ will be :

- (A) 0.44×10^{15}
- (B) 6.57×10^{15}
- (C) 4.38×10^{15}
- (D) 2.46×10^{15}

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(D) 2.46×10^{15}

Solution:

$$nf_1 = k \left(\frac{1}{1} - \frac{1}{3^2} \right)$$

$$nf_2 = k \left(1 - \frac{1}{2^2} \right)$$

$$\frac{f_1}{f_2} = \frac{8/9}{3/4} \Rightarrow f_2 = 2.46 \times 10^{15}$$

Q11: In a photoelectric experiment ultraviolet light of wavelength 280 nm is used with lithium cathode having work function $\phi = 2.5 \text{ eV}$. If the wavelength of incident light is switched to 400 nm, find out the change in the stopping potential.

$(h = 6.63 \times 10^{-34} \text{ Js}, c = 3 \times 10^8 \text{ ms}^{-1})$

- (A) 1.3 V
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- (C) 1.9 V
- (D) 0.6 V

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Solution:

$$KE_{\text{max}} = eV_s = \frac{hc}{\lambda} - \phi$$

$$\Rightarrow eV_{s_1} = \frac{1240}{280} - 2.5 = 1.93 \text{ eV}$$

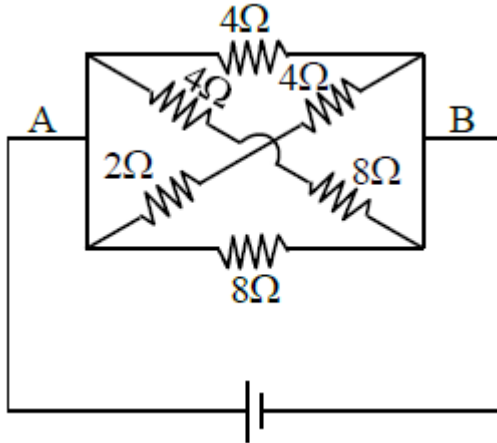
$$\rightarrow V_{s_1} = 1.93 \text{ V} \dots (i)$$

$$\rightarrow eV_{s_2} = \frac{1240}{400} - 2.5 = 0.6 \text{ eV}$$

$$\Rightarrow V_{s_2} = 0.6 \text{ V} \dots (ii)$$

$$\Delta V = V_{s_1} - V_{s_2} = 1.93 - 0.6 = 1.33 \text{ V}$$

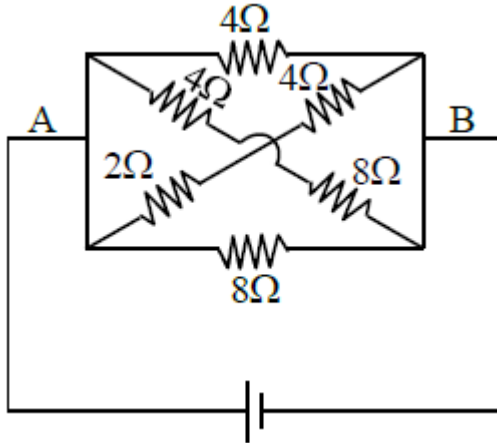
Q12: In the given figure, the emf of the cell is 2.2 V and if internal resistance is $0.6\ \Omega$. Calculate the power dissipated in the whole circuit:



2.2 V , $r = 0.6\ \Omega$

- (A) 1.32 W
- (B) 0.65 W
- (C) 2.2 W
- (D) 4.4 W

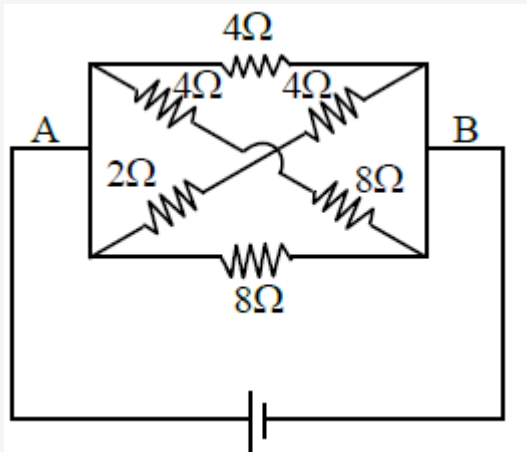
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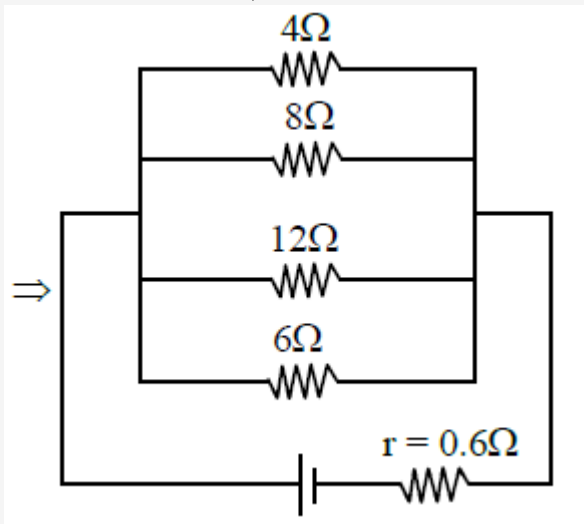
$2.2\text{ V}, r = 0.6\ \Omega$

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$2.2\text{ V}, r = 0.6\Omega$

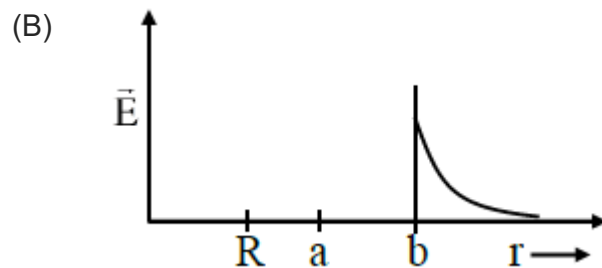
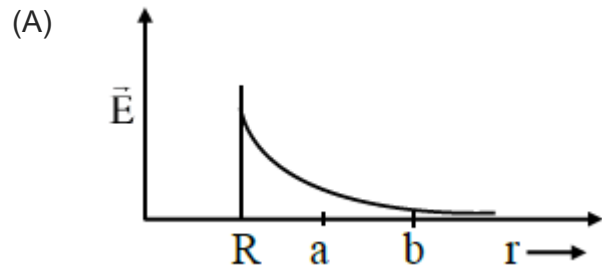
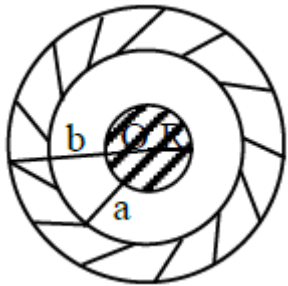


$$\frac{1}{R_{eq}} = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \frac{1}{6} = \frac{6+3+2+4}{24} = \frac{15}{24}$$

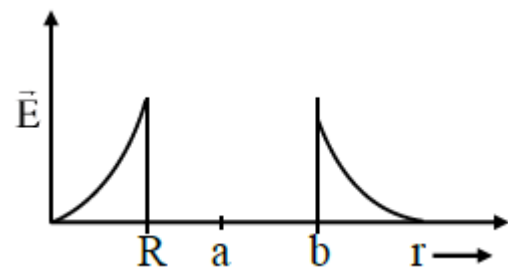
$$R_{eq} = \frac{24}{15} = 1.6 \Rightarrow R_T = 1.6 + 0.6 = 2.2\Omega$$

$$P = \frac{V^2}{R_T} = \frac{(2.2)^2}{2.2} = 2.2W$$

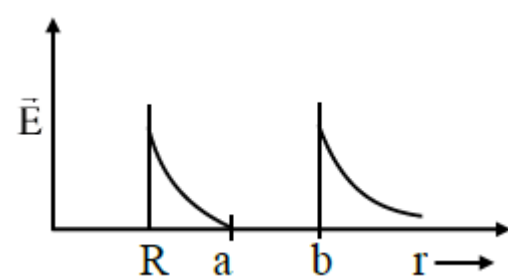
Q13: A solid metal sphere of radius R having charge q is enclosed inside the concentric spherical shell of inner radius a and outer radius b as shown in figure. The approximate variation electric field $\vec{E}$ as a function of distance r from centre O is given by



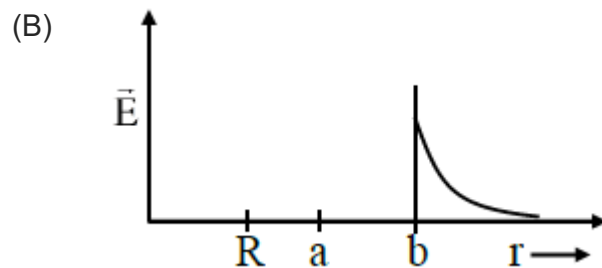
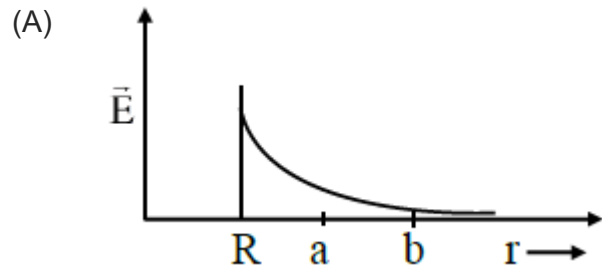
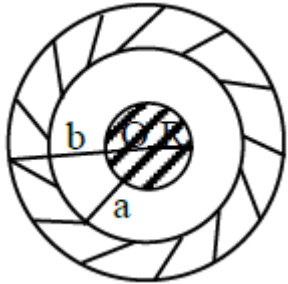
(C)



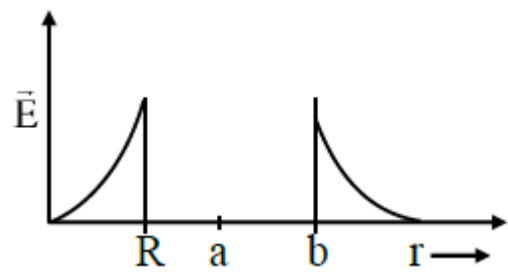
(D)



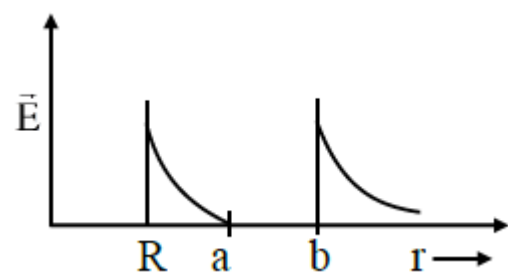
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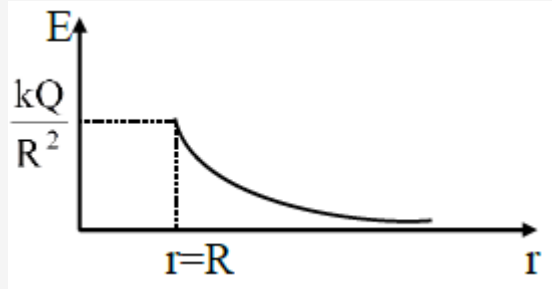


Solution:

Considering outer spherical shell is non-conducting
Electric field inside a metal sphere is zero.

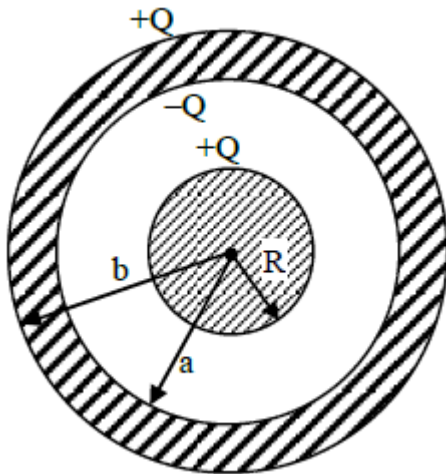
$$r < R \Rightarrow E = 0$$

$$r > R \Rightarrow E = \frac{kQ}{r^2}$$



Option (b)

Considering outer spherical shell is conducting

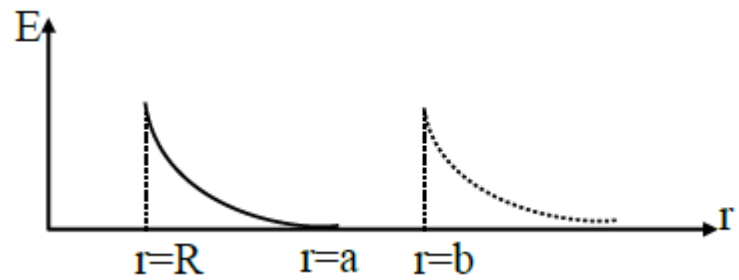


$$r < R, \quad E = 0$$

$$R \leq r < a \quad E = \frac{kQ}{r^2}$$

$$a \leq r < b, \quad E = 0$$

$$r \geq b \quad E = \frac{kQ}{r^2}$$



Q14: The rms speeds of the molecules of Hydrogen, Oxygen and Carbon dioxide at the same temperature are V_H , V_O and V_C respectively then:

(A) $V_H > V_O > V_C$

(B) $V_C > V_O > V_H$

(C) $V_H = V_O > V_C$

(D) $V_H = V_O = V_C$

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(D) $V_H = V_O = V_C$

Solution:

$$V_{RMS} = \sqrt{\frac{3RT}{M_w}}$$

At the same temperature $V_{RMS} \propto \frac{1}{\sqrt{M_w}}$

$$\Rightarrow V_H > V_O > V_C$$

Q15: In a Screw Gauge, fifth division of the circular scale coincides with the reference line when the ratchet is closed. There are 50 divisions on the circular scale, and the main scale moves by 0.5 mm on a complete rotation. For a particular observation the reading on the main scale is 5 mm and the 20th division of the circular scale coincides with reference line. Calculate the true reading.

- (A) 5.00 mm
- (B) 5.25 mm
- (C) 5.15 mm
- (D) 5.20 mm

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Solution:

$$\text{Least count (L.C)} = \frac{0.5}{50}$$

$$\text{True reading} = 5 + \frac{0.5}{50} \times 20 - \frac{0.5}{50} \times 5$$

$$= 5 + \frac{0.5}{50}(15) = 5.15 \text{ mm}$$

Q16: What equal length of an iron wire and a copper-nickel alloy wire, each of 2 mm diameter connected parallel to give an equivalent resistance of 3Ω ?

(Given resistivities of iron and copper-nickel alloy wire are $12 \mu\Omega \text{ cm}$ and $51 \mu\Omega \text{ cm}$ respectively)

- (A) 82 m
- (B) 97 m
- (C) 110 m
- (D) 90 m

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- (A) 82 m
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- (D) 90 m

Solution:

$$\frac{R_1 R_2}{R_1 + R_2} = 3$$

$$\frac{\frac{(12 \times 10^{-6} \times 10^{-2}) \ell \times 4}{\pi (2)^2 \times 10^{-6}} \times \frac{(51 \times 10^{-6} \times 10^{-2}) \ell \times 4}{\pi (2)^2 \times 10^{-6}}}{\frac{63 \times 10^{-6} \times 10^{-2} \times \ell \times 4}{\pi (2)^2 \times 10^{-6}}} = 3$$

$$\Rightarrow \ell = 97m$$

Q17: The initial mass of a rocket is 1000 kg. Calculate at what rate the fuel should be burnt so that the rocket is given an acceleration of 20 ms^{-2} . The gases come out at a relative speed of 500 ms^{-1} with respect to the rocket : [Use $g = 10 \text{ m/s}^2$]

- (A) $6.0 \times 10^2 \text{ kg s}^{-1}$
- (B) 500 kg s^{-1}
- (C) 10 kg s^{-1}
- (D) 60 kg s^{-1}

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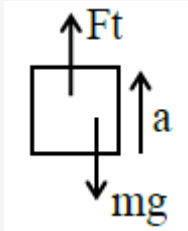
(A) $6.0 \times 10^2 \text{ kg s}^{-1}$

(B) 500 kg s^{-1}

(C) 10 kg s^{-1}

(D) 60 kg s^{-1}

Solution:



$$F_{thrust} = \left(\frac{dm}{dt} \cdot V_{rel} \right)$$

$$\left(\frac{dm}{dt} V_{rel} - mg \right) = ma$$

$$\Rightarrow \left(\frac{dm}{dt} \right) \times 500 - 10^3 \times 10 = 10^3 \times 20$$

$$\frac{dm}{dt} = (60 \text{ kg/s})$$

Q18: If E, L, M and G denote the quantities as energy, angular momentum, mass and constant of gravitation respectively, then the dimensions of P in the formula

$P = EL^2M^{-5}G^{-2}$ are :-

- (A) $[M^0L^1T^0]$
- (B) $[M^{-1}L^{-1}T^2]$
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Solution:

$$E = ML^2T^{-2}$$

$$L = ML^2T^{-1}$$

$$m = M$$

$$G = M^{-1}L^3T^{-2}$$

$$P = \frac{EL^2}{M^5G^2}$$

$$[P] = \frac{(ML^2T^{-2})(M^2L^4T^{-2})}{M^5(M^{-2}L^6T^{-4})} = M^0L^0T^0$$

Q19: The material filled between the plates of a parallel plate capacitor has resistivity $200 \, \Omega m$. The value of capacitance of the capacitor is 2 pF. If a potential difference of 40 V is applied across the plates of the capacitor, then the value of leakage current flowing out of the capacitor is : (given the value of relative permittivity of material is 50)

- (A) $9.0 \, \mu A$
- (B) $9.0 \, mA$
- (C) $0.9 \, mA$
- (D) $0.9 \, \mu A$

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- (B) $9.0 \, mA$
- (C) $0.9 \, mA$
- (D) $0.9 \, \mu A$

Solution:

$$\rho = 200 \, \Omega m$$

$$C = 2 \times 10^{-12} F$$

$$V = 40 V$$

$$K = 50$$

$$i = \frac{q}{\rho k \epsilon_0} = \frac{q_0}{\rho k \epsilon_0} e^{-\frac{t}{\rho k \epsilon_0}}$$

$$i_{\max} = \frac{2 \times 10^{-12} \times 40}{200 \times 50 \times 8.85 \times 10^{-12}}$$

$$= \frac{80}{10^4 \times 8.85} = 903 \, \mu A = 0.9 mA$$

Q20: **Statement-I** : By doping silicon semiconductor with pentavalent material, the electrons density increases.

Statement-II : The n-type semiconductor has net negative charge.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Statement-I is true but Statement-II is false.
- (B) Statement-I is false but Statement-II is true.
- (C) Both Statement-I and Statement-II are true.
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Solution:

Pentavalent activities have excess free e^-

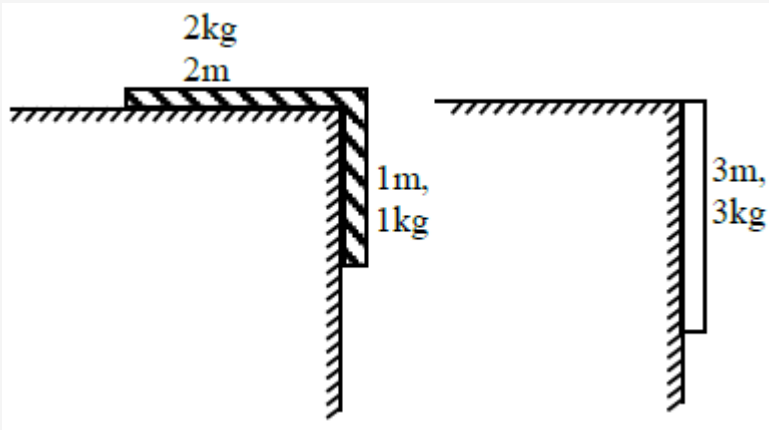
So e^- density increases but overall semiconductor is neutral.

Q21: A uniform chain of length 3 meters and mass 3 kg overhangs a smooth table with 2 meters laying on the table. If k is the kinetic energy of the chain in joule as it completely slips off the table, then the value of k is (Take $g = 10\text{m/s}^2$)

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40.00

Solution:



From energy conservation

$$K_i + U_i = K_f + U_f$$

$$0 + \left(-1 \times 10 \times \frac{1}{2} \right) = K_f + \left(-3 \times 10 \times \frac{3}{2} \right)$$

$$-5 = K_f - 45$$

$$K_f = 40 \text{ J}$$

Ans. 40.00

Q22: The electric field in a plane electromagnetic wave is given by

$$\vec{E} = 200 \cos \left[\left(\frac{0.5 \times 10^3}{m} \right) x - \left(1.5 \times 10^{11} \frac{rad}{s} \times t \right) \right] \frac{V}{m} \hat{j}$$

If this wave falls normally on a perfectly reflecting surface having an area of 100 cm^2 . If the radiation pressure exerted by the E.M. wave on the surface during a 10 minute exposure is $\frac{x}{10^9} \frac{N}{m^2}$. Find the value of x.

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354.00

Solution:

$$E_0 = 200$$

$$I = \frac{1}{2} \epsilon_0 E_0^2 \cdot C$$

Radiation pressure

$$P = \frac{2I}{C}$$

$$= \left(\frac{2}{C} \right) \left(\frac{1}{2} \epsilon_0 E_0^2 C \right)$$

$$= \epsilon_0 E_0^2$$

$$= 8.85 \times 10^{-12} \times 200^2$$

$$= 8.85 \times 10^{-8} \times 4$$

$$= \frac{354}{10^9}$$

Ans. 354.0

Q23: A source and a detector move away from each other in absence of wind with a speed of 20 m/s with respect to the ground. If the detector detects a frequency of 1800 Hz of the sound coming from the source, then the original frequency of source considering speed of sound in air 340 m/s will be Hz.

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2025.00

Solution:



$$V_S = 20 \text{ m/s}$$

$$V_O = 20 \text{ m/s}$$

$$f' = f \left(\frac{C - V_O}{C + V_S} \right)$$

$$1800 = f \left(\frac{340 - 20}{340 + 20} \right)$$

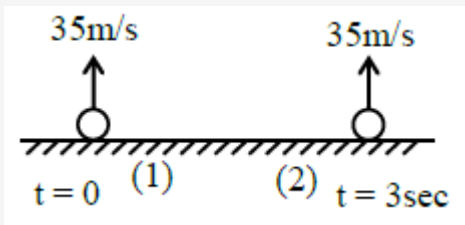
$$f = 2025 \text{ Hz}$$

Ans. 2025

Q24: Two spherical balls having equal masses with radius of 5 cm each are thrown upwards along the same vertical direction at an interval of 3s with the same initial velocity of 35 m/s, then these balls collide at a height ofm. (Take $g = 10 \text{ m/s}^2$)

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Solution:



When both balls will collide

$$y_1 = y_2$$

$$35t - \frac{1}{2} \times 10 \times t^2 = 35(t - 3) - \frac{1}{2} \times 10 \times (t - 3)^2$$

$$35t - \frac{1}{2} \times 10 \times t^2 = 35t - 105 - \frac{1}{2} \times 10 \times t^2$$

$$-\frac{1}{2} \times 10 \times 3^2 + \frac{1}{2} \times 10 \times 6t$$

$$0 = 150 - 30t$$

$$t = 5 \text{ sec}$$

∴ Height at which both balls will collide

$$h = 35t - \frac{1}{2} \times 10 \times t^2$$

$$= 35 \times 5 - \frac{1}{2} \times 10 \times 5^2$$

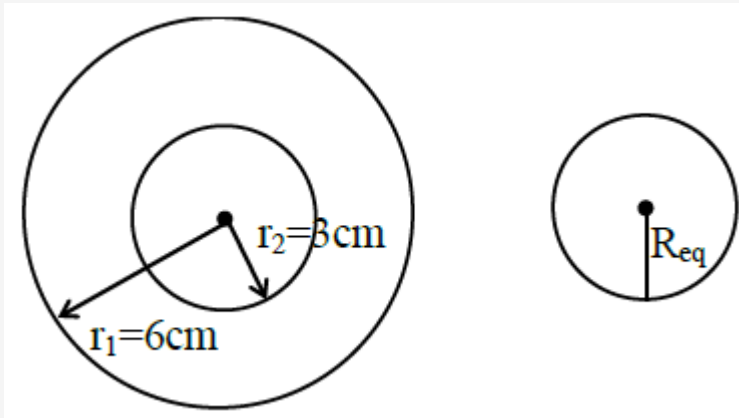
$$h = 50 \text{ m}$$

Ans. 50.00

Q25: A soap bubble of radius 3 cm is formed inside the another soap bubble of radius 6 cm. The radius of an equivalent soap bubble which has the same excess pressure as inside the smaller bubble with respect to the atmospheric pressure is cm.

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Solution:



Excess pressure inside the smaller soap bubble

$$\Delta P = \frac{4S}{r_1} + \frac{4S}{r_2} \dots (i)$$

The excess pressure inside the equivalent soap bubble

$$\Delta P = \frac{4S}{R_{eq}} \dots (ii)$$

From (i) & (ii)

$$\frac{4S}{R_{eq}} = \frac{4S}{r_1} + \frac{4S}{r_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$$

$$= \frac{1}{6} + \frac{1}{3}$$

$$R_{eq} = 2 \text{ cm}$$

Ans. 2.00

Q26: An amplitude modulated wave is represented by

$C_m(t) = 10(1 + 0.2 \cos 12560t) \sin(111 \times 10^4 t)$ volts. The modulating frequency in kHz will be

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Solution:

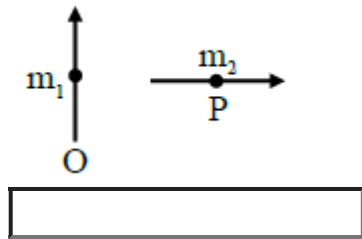
$$W_m = 12560 = 2\pi f_m$$

$$f_m = \frac{12560}{2\pi}$$

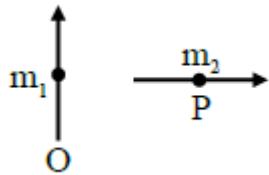
$$= 2000 \text{ Hz}$$

Ans. 2.00

Q27: Two short magnetic dipoles m_1 and m_2 each having magnetic moment of 1 Am^2 are placed at point O and P respectively. The distance between OP is 1 meter. The torque experienced by the magnetic dipole m_2 due to the presence of m_1 is $\dots \times 10^{-7} \text{ Nm}$.

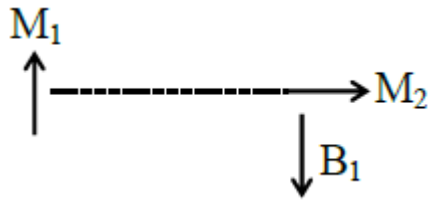


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1.00

Solution:



$$\begin{aligned}\vec{\tau} &= \vec{M}_2 \times \vec{B}_1 \\ \tau &= M_2 B_1 \sin 90^\circ \\ &= 1 \times \frac{\mu_0}{4\pi} \frac{M_1}{(1)^3} 1 \\ &= 10^{-7} \text{ N.m}\end{aligned}$$

Ans. 1.00

Q28: Two travelling waves produces a standing wave represented by equation,
 $y = 1.0 \text{ mm} \cos(1.57 \text{ cm}^{-1}) x \sin(78.5 \text{ s}^{-1}) t$. The node closest to the origin in the region $x > 0$ will be at $x = \dots\dots\dots \text{cm}$.

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Solution:

For node

$$\cos(1.57 \text{ cm}^{-1}) x = 0$$

$$(1.57 \text{ cm}^{-1}) x = \frac{\pi}{2}$$

$$x = \frac{\pi}{2(1.57)} \text{ cm} = 1 \text{ cm}$$

Ans. 1.00

Q29: White light is passed through a double slit and interference is observed on a screen 1.5 m away. The separation between the slits is 0.3 mm. The first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central white fringes. The difference in wavelengths of red and violet light is nm.

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300.00

Solution:

Position of bright fringe $y = n \frac{D\lambda}{d}$

$$y_1 \text{ of red} = \frac{D\lambda_r}{d} = 3.5 \text{ mm}$$

$$\lambda_r = 3.5 \times 10^{-3} \frac{d}{D}$$

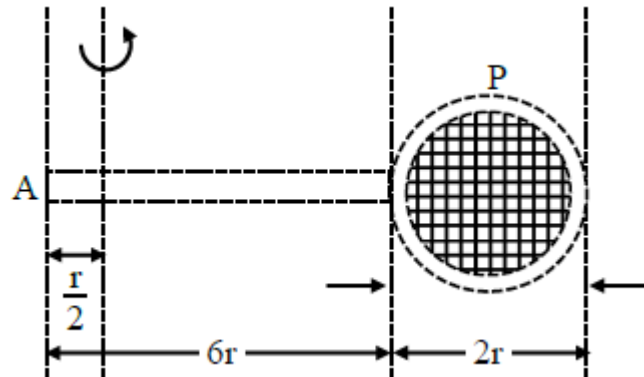
$$\text{Similarly } \lambda_v = 2 \times 10^{-3} \frac{d}{D}$$

$$\lambda_r - \lambda_v = (1.5 \times 10^{-3}) \left(\frac{0.3 \times 10^{-3}}{1.5} \right)$$

$$= 3 \times 10^{-7} = 300 \text{ nm}$$

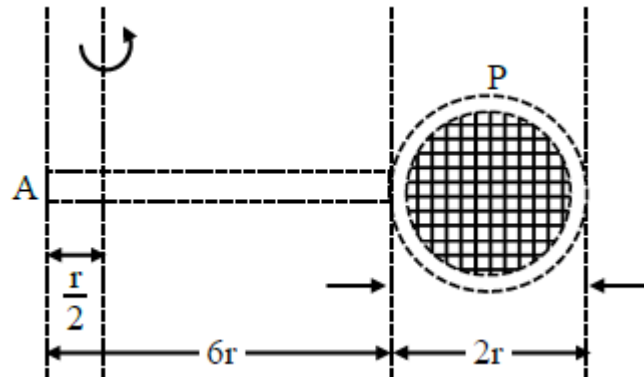
Ans. 300.0

Q30: Consider a badminton racket with length scales as shown in the figure.



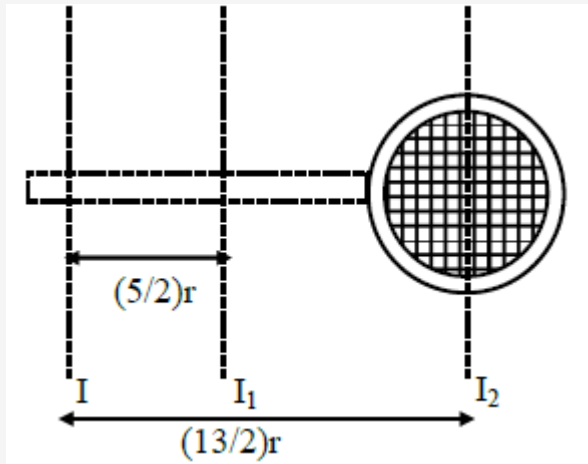
If the mass of the linear and circular portions of the badminton racket are same (M) and the mass of the threads are negligible, the moment of inertia of the racket about an axis perpendicular to the handle and in the plane of the ring at, $\frac{r}{2}$ distance from the end A of the handle will be Mr^2 .

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If the mass of the linear and circular portions of the badminton racket are same (M) and the mass of the threads are negligible, the moment of inertia of the racket about an axis perpendicular to the handle and in the plane of the ring at, $\frac{r}{2}$ distance from the end A of the handle will be Mr^2 .

Solution:



$$\begin{aligned}
 I &= \left[I_1 + M \left(\frac{5}{2}r \right)^2 \right] + \left[I_2 + M \left(\frac{13r}{2} \right)^2 \right] \\
 &= \left[\frac{m (36r^2)}{12} + \frac{M (25r^2)}{4} \right] + \left[\frac{Mr^2}{2} + \frac{169Mr^2}{4} \right] \\
 &= 52 Mr^2
 \end{aligned}$$

Ans. 52.00

Q31: Which of the following complexes is violet in colour?

- (A) $[\text{Fe}(\text{CN})_6]^{4-}$
- (B) $[\text{Fe}(\text{SCN})_6]^{4-}$
- (C) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot \text{H}_2\text{O}$
- (D) $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$

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Solution:

- (a) $[\text{Fe}(\text{CN})_6]^{4-} \rightarrow$ Pale yellow solution
- (b) $[\text{Fe}(\text{SCN})_6]^{4-} \rightarrow$ Blood red colour
- (c) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot \text{H}_2\text{O} \rightarrow$ Prussian blue
- (d) $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-} \rightarrow$ Violet colour

Q32: Which one of the following is correct for the adsorption of a gas at a given temperature on a solid surface?

- (A) $\Delta H > 0, \Delta S > 0$
- (B) $\Delta H > 0, \Delta S < 0$
- (C) $\Delta H < 0, \Delta S < 0$
- (D) $\Delta H < 0, \Delta S > 0$

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Solution:

During adsorption, there is always a decrease in residual forces of the surface, i.e., there is decrease in surface energy which appears as heat. Adsorption, therefore, is invariably an exothermic process. In other words, ΔH of adsorption is always negative.

When a gas is adsorbed, the freedom of movement of its molecules become restricted. This amounts to decrease in the entropy of the gas after adsorption, i.e., ΔS is negative.

Adsorption is thus accompanied by decrease in enthalpy as well as decrease in entropy of the system.

Q33: Which one of the following when dissolved in water gives coloured solution in nitrogen atmosphere?

- (A) CuCl_2
- (B) AgCl
- (C) ZnCl_2
- (D) Cu_2Cl_2

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Solution:

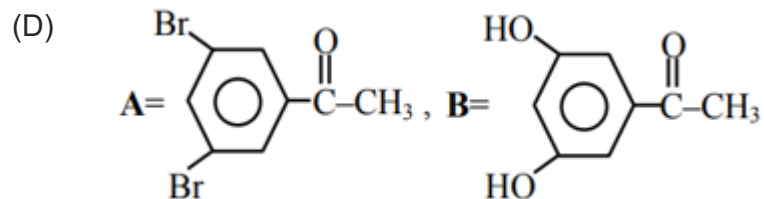
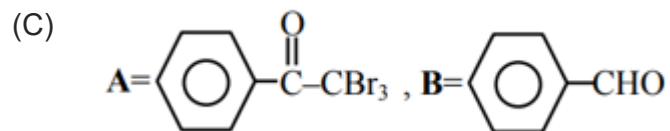
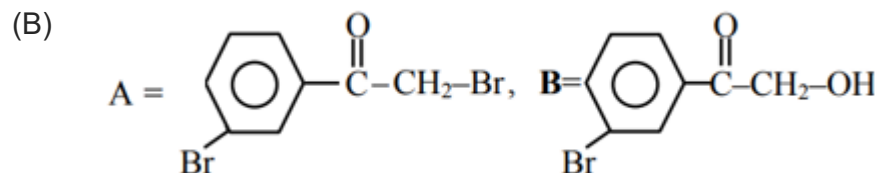
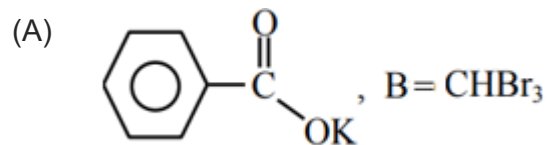
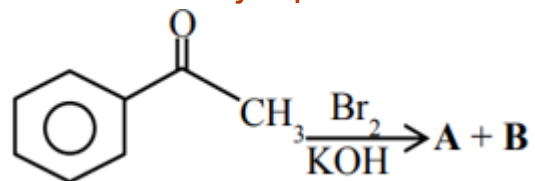
(a) $\text{CuCl}_2 + n\text{H}_2\text{O} \rightarrow \text{Cu}_{(\text{aq.})}^{+2}$ (blue colour)

(b) $\text{AgCl} + n\text{H}_2\text{O} \rightarrow \text{Insoluble}$

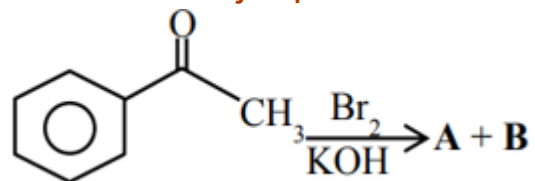
(c) $\text{ZnCl}_2 + n\text{H}_2\text{O} \rightarrow \text{Zn}_{(\text{aq.})}^{+2}$ (Colourless)

(d) $\text{Cu}_2\text{Cl}_2 + n\text{H}_2\text{O} \rightarrow \text{Insoluble}$

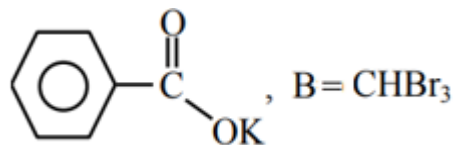
Q34: The major product formed in the following reaction sequence A and B are:



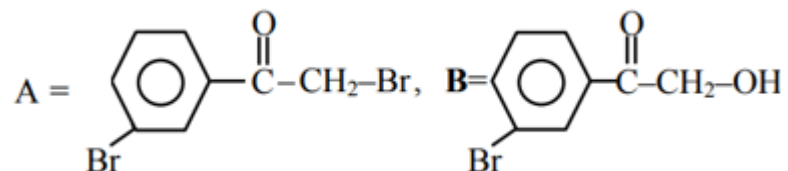
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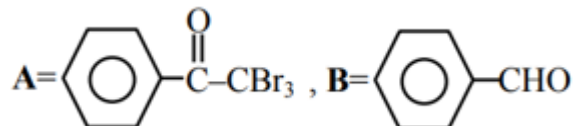
(A)



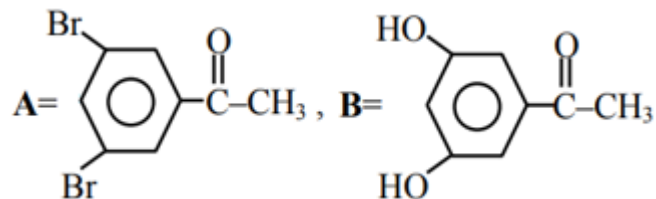
(B)



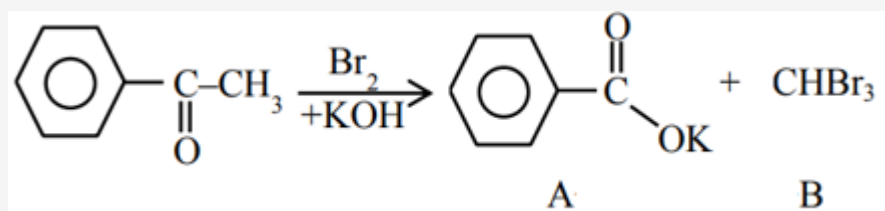
(C)



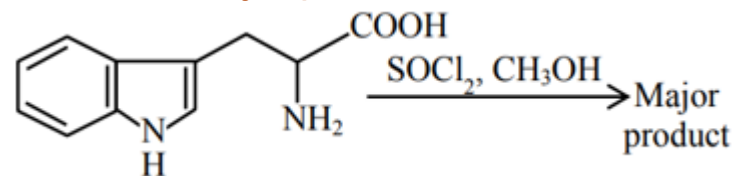
(D)

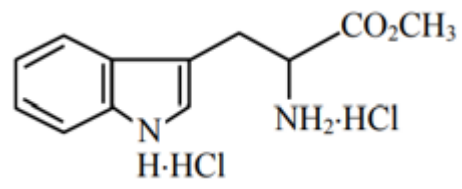
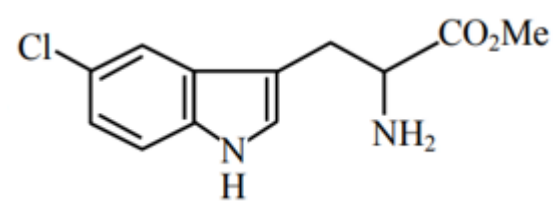
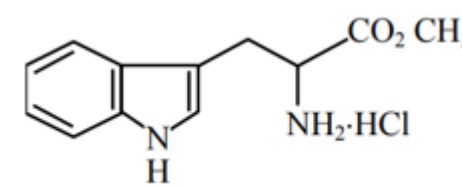
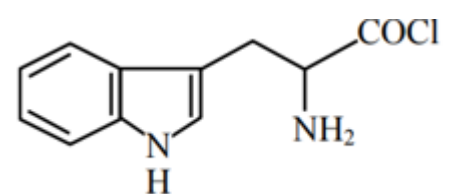


Solution:

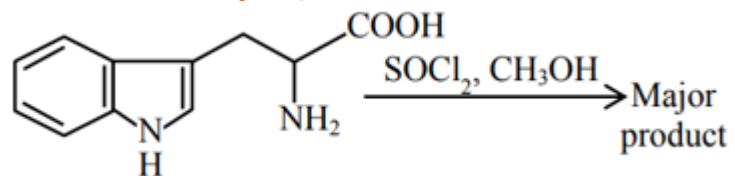


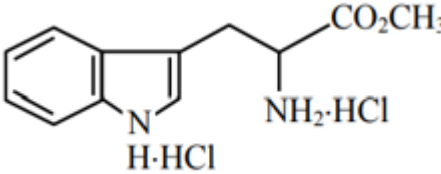
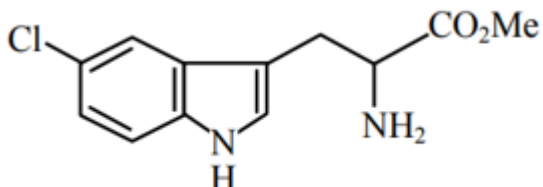
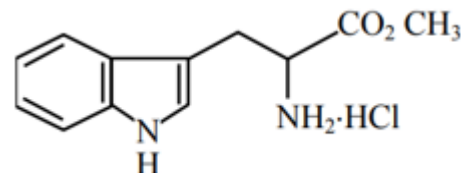
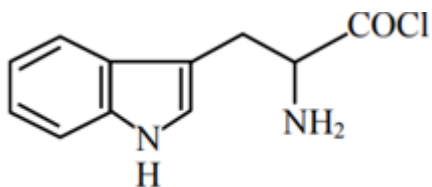
Q35: The major product formed in the following reaction is:



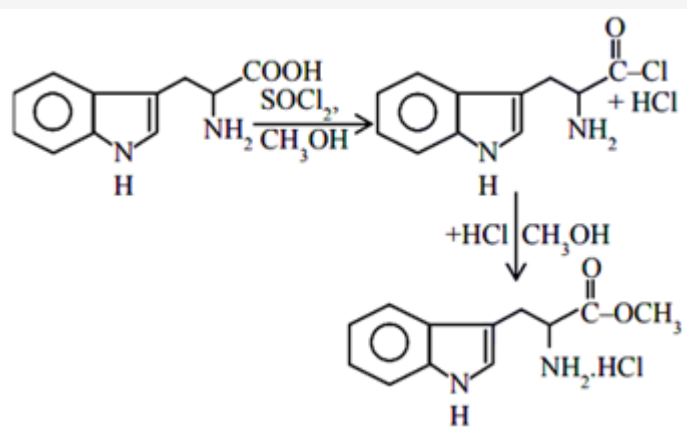
- (A) 
Structure: Indole-3-ethyl-L-tryptophan methyl ester hydrochloride salt. The indole ring is attached to a 2-amino-3-methoxycarbonylpropyl side chain. The amino group is shown as $\text{NH}_2 \cdot \text{HCl}$ and the carboxyl group is shown as CO_2CH_3 .
- (B) 
Structure: 5-chloro-L-tryptophan methyl ester. The indole ring has a chlorine atom at the 5-position and is attached to a 2-amino-3-methoxycarbonylpropyl side chain. The amino group is shown as NH_2 and the carboxyl group is shown as CO_2Me .
- (C) 
Structure: L-tryptophan methyl ester hydrochloride salt. The indole ring is attached to a 2-amino-3-methoxycarbonylpropyl side chain. The amino group is shown as $\text{NH}_2 \cdot \text{HCl}$ and the carboxyl group is shown as CO_2CH_3 .
- (D) 
Structure: L-tryptophan acid chloride. The indole ring is attached to a 2-amino-3-chloroacetylpropyl side chain. The amino group is shown as NH_2 and the carboxyl group is shown as COCl .

Q35: The major product formed in the following reaction is:

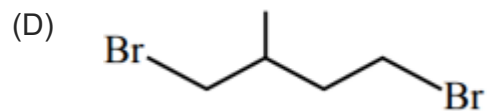
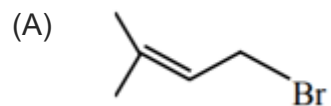
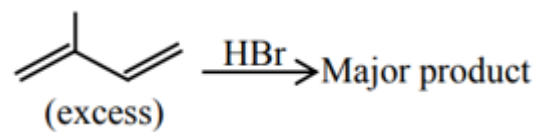


- (A) 
- (B) 
- (C) 
- (D) 

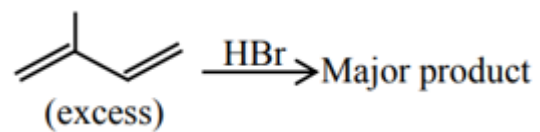
Solution:



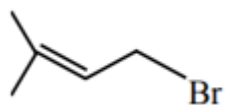
Q36: The major product formed in the following reaction is:



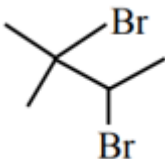
Q36: The major product formed in the following reaction is:



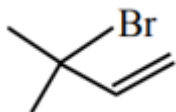
(A)



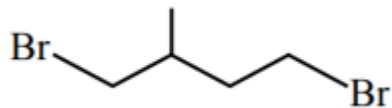
(B)



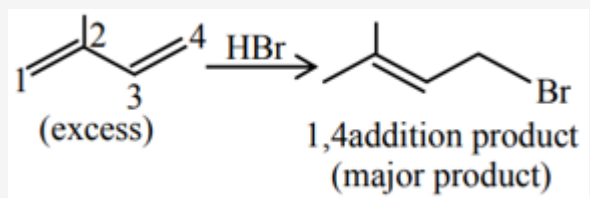
(C)



(D)



Solution:



Q37: The polymer formed on heating Novolac with formaldehyde is:

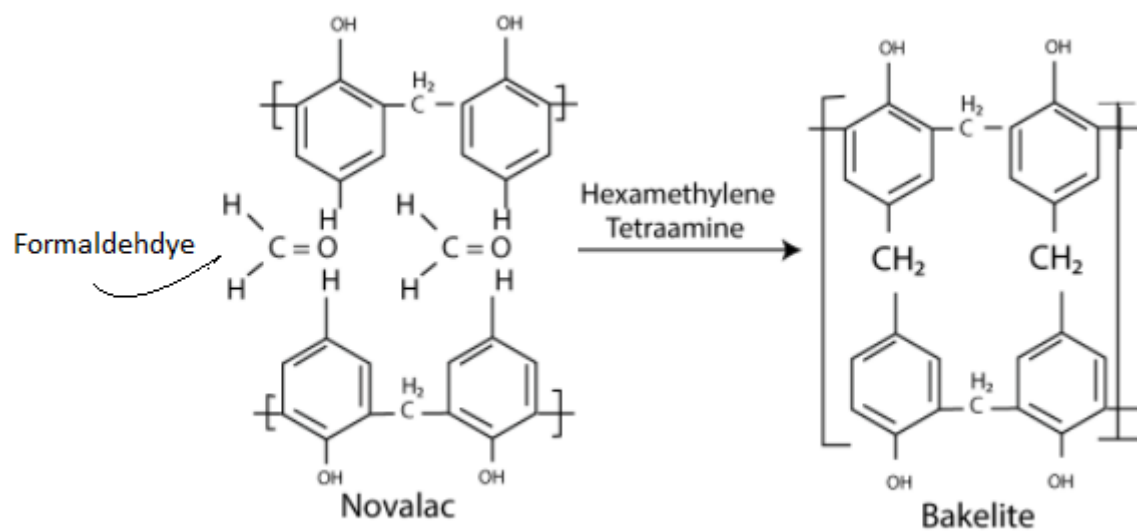
- (A) Bakelite
- (B) Polyester
- (C) Melamine
- (D) Nylon

Q37: The polymer formed on heating Novolac with formaldehyde is:

- (A) Bakelite
- (B) Polyester
- (C) Melamine
- (D) Nylon

Solution:

Novolac + formaldehyde $\rightarrow$ Bakelite



Q38: Given below are two statements:

Statement I: The limiting molar conductivity of KCl (strong electrolyte) is higher compared to that of CH_3COOH (weak electrolyte).

Statement II: Molar conductivity decreases with decrease in concentration of electrolyte.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

- (A) **Statement I** is true, but **Statement II** is false
- (B) **Statement I** is false, but **Statement II** is true
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- (D) Both **Statement I** and **Statement II** are false

Solution:

According to NCERT data,

Ion	H ⁺	K ⁺	Cl ⁻	CH ₃ COO ⁻
$\Lambda_m^\circ \text{ (Scm}^2 \text{ / mole)}$	349.6	73.5	76.3	40.9

Now,

$$\Lambda_m^\circ (\text{CH}_3\text{COOH}) = \Lambda_m^\circ (\text{H}^+) + \Lambda_m^\circ (\text{CH}_3\text{COO}^-)$$

$$\Lambda_m^\circ (\text{CH}_3\text{COOH}) = 349.6 + 40.9$$

$$\Lambda_m^\circ (\text{CH}_3\text{COOH}) = 390.5 \text{ Scm}^2/\text{mole}$$

And,

$$\Lambda_m^\circ (\text{KCl}) = \Lambda_m^\circ (\text{K}^+) + \Lambda_m^\circ (\text{Cl}^-)$$

$$\Lambda_m^\circ (\text{KCl}) = 73.5 + 76.3$$

$$\Lambda_m^\circ (\text{KCl}) = 149.8 \text{ Scm}^2/\text{mole}$$

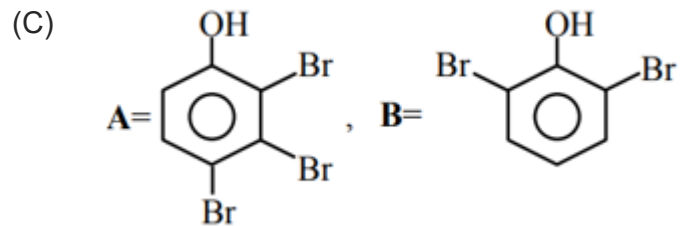
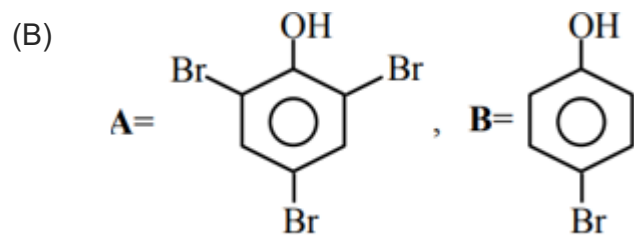
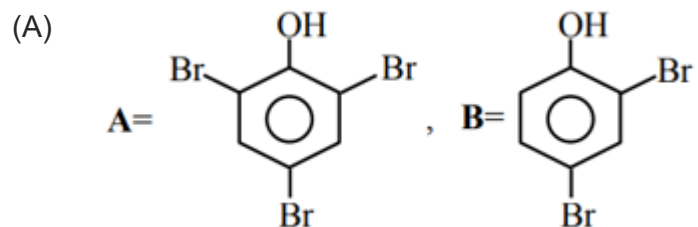
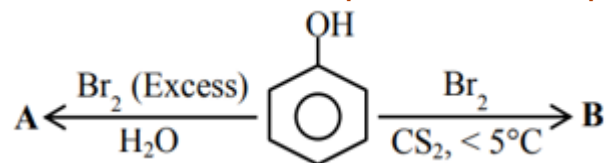
So, statement-I is wrong or False.

$$\Lambda_m = kV$$

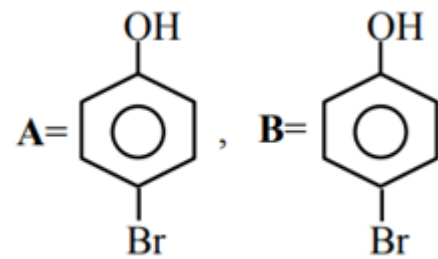
Molar conductivity increases with decrease in concentration. This is because the total volume, V, of solution containing one mole of electrolyte also increases. It has been found that decrease in κ on dilution of a solution is more than compensated by increase in its volume.

So, statement-II is also wrong or False.

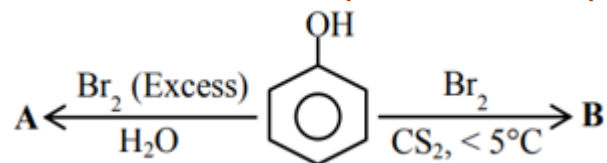
Q39: The correct options for the products A and B of the following reactions are:



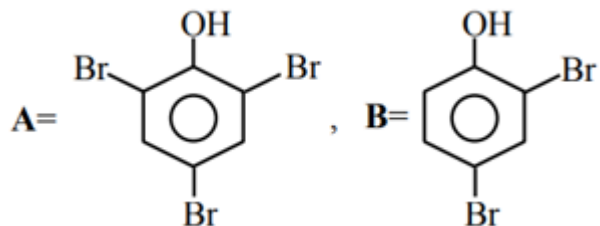
(D)



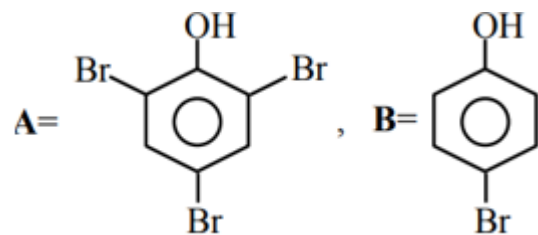
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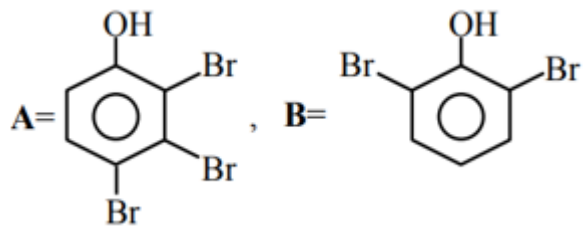
(A)



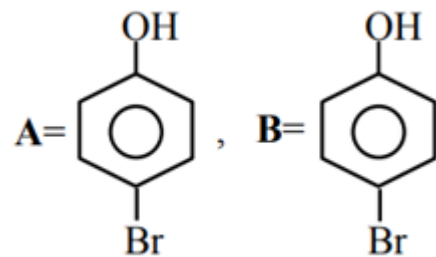
(B)



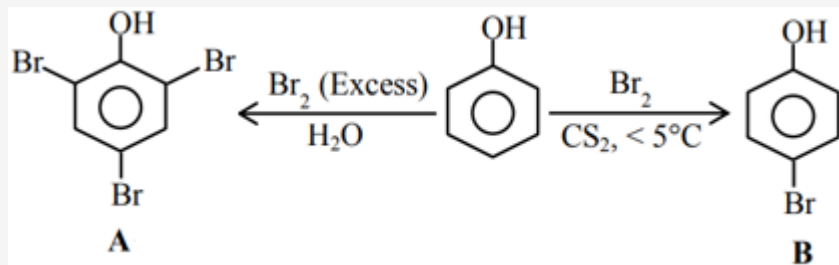
(C)



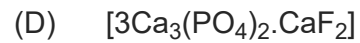
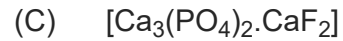
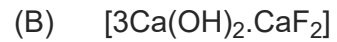
(D)



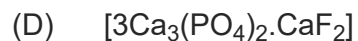
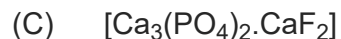
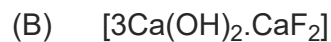
Solution:



Q40: The conversion of hydroxyapatite occurs due to presence of F^- ions in water. The correct formula of hydroxyapatite is:



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Solution:

The F^- ions make the enamel on teeth much harder by converting hydroxyapatite, $[3(Ca_3(PO_4)_2) \cdot Ca(OH)_2]$, the enamel on the surface of the teeth into much harder fluorapatite, $[3Ca_3(PO_4)_2 \cdot CaF_2]$.

Q41: Given below are two statements.

Statement I: In the titration between strong acid and weak base methyl orange is suitable as an indicator.

Statement II: For titration of acetic acid with NaOH phenolphthalein is not a suitable indicator.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

- (A) **Statement I** is false, but **Statement II** is true
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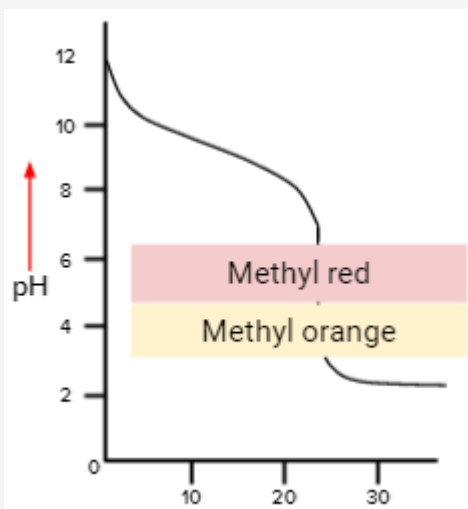
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Solution:

For titration of **weak base-strong acid**, the vertical line generally extends from $\text{pH} = 3.5$ to $\text{pH} = 6$. Thus, good choices would be-

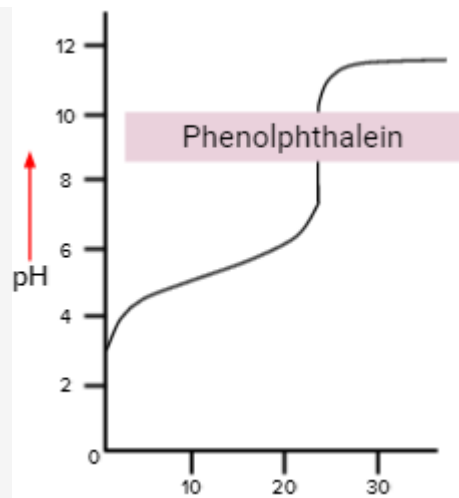
- Methyl orange (3.2 - 4.4)
- Methyl red (4.2 - 6.3)



Thus, statement I is true.

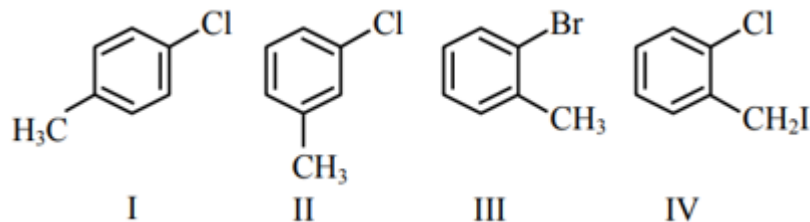
For titration of acetic acid (weak acid) with NaOH (strong base) the vertical line extends from $\text{pH} = 8$ to $\text{pH} = 10$. Thus, good choice would be-

- Phenolphthalein (8.2 - 10)



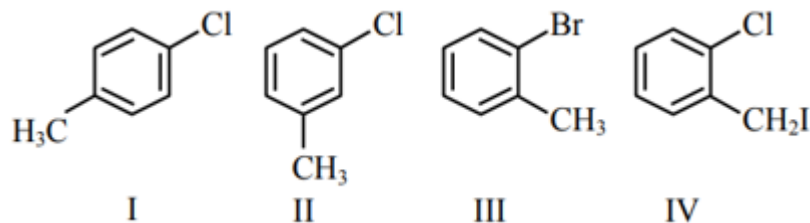
Thus, statement II is false.

Q42: Among the following compounds I-IV, which one forms a yellow precipitate on reacting sequentially with (i) NaOH (ii) dil. HNO_3 (iii) AgNO_3 ?



- (A) II
- (B) IV
- (C) I
- (D) III

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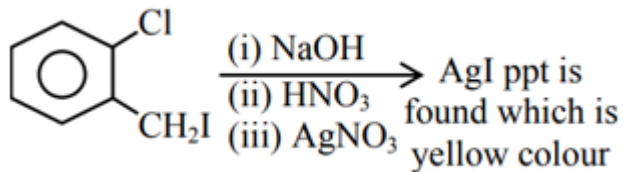
(B) IV

(C) I

(D) III

Solution:

Silver nitrate test is carried out to detect the type of halide ion in a given solution.



Other compounds halide can't be removed because corresponding C^+ is highly unstable.

Q43: Which one of the following methods is most suitable for preparing deionized water?

- (A) Synthetic resin method
- (B) Clark's method
- (C) Calgon's method
- (D) Permutit method

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Solution:

Pure demineralized (de-ionized) water free from all soluble mineral salts is obtained by passing water successively through a cation exchange (in the H^+ form) and an anion exchange (in the OH^- form) resins.

Q44: Given below are two statements.

Statement I: The choice of reducing agents for metals extraction can be made by using Ellingham diagram, a plot of ΔG vs temperature.

Statement II: The value of ΔS increases from left to right in Ellingham diagram.

In the light of the above statements, choose statements, choose the **most appropriate** answer from the options given below.

- (A) Both **Statement I** and **Statement II** are true
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Solution:

Given statement-I is true as in a number of processes, one element is used to reduce the oxide of another metal. Any element will reduce the oxide of other metal which lie above it in the Ellingham diagram because the free energy change will become more negative.

Given statement -II is false as the value of ΔS increases from left to right only when there is a phase change otherwise it remains constant.

Q45: What are the products formed in sequence when excess of CO_2 is passed in slaked lime?

- (A) $\text{Ca}(\text{HCO}_3)_2$, CaCO_3
- (B) CaCO_3 , $\text{Ca}(\text{HCO}_3)_2$
- (C) CaO , $\text{Ca}(\text{HCO}_3)_2$
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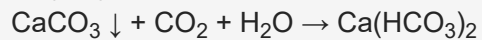
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Solution:



Q46: Given below are two statements.

Statement I: According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increase with decrease in positive charges on the nucleus as there is no strong hold on the electron by the nucleus.

Statement II: According Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in principal quantum number.

In the light of the above statements, choose the **most appropriate answer from the options given below:**

- (A) Both **Statement I** and **Statement II** are false
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Solution:

Velocity of electron in Bohr's atom is given by

$$v \propto \frac{Z}{n}$$

Z = atomic number of atom, corresponds to +ve charge on nucleus. So, as Z increases, velocity increases. So, statement-I is wrong.

And as ' n ' decreases, velocity increase. So, statement-II is correct.

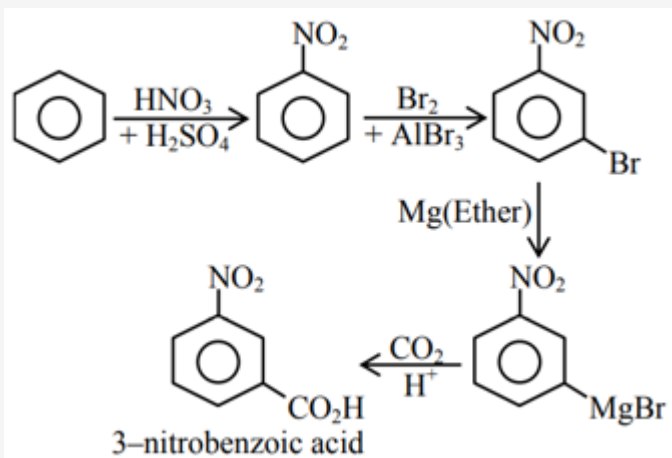
Q47: The correct sequential addition of reagents in the preparation of 3-nitrobenzoic acid from benzene is:

- (A) $\text{Br}_2/\text{AlBr}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$, Mg/ether , CO_2 , H_3O^+
- (B) $\text{Br}_2/\text{AlBr}_3$, NaCN , H_3O^+ , $\text{HNO}_3/\text{H}_2\text{SO}_4$
- (C) $\text{Br}_2/\text{AlBr}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$, NaCN , CO_2 , H_3O^+
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- (D) $\text{HNO}_3/\text{H}_2\text{SO}_4$, $\text{Br}_2/\text{AlBr}_3$, Mg/ether , CO_2 , H_3O^+

Solution:



Q48: Given below are two statements.

Statement I: Frenkel defects are vacancy as well as interstitial defects.

Statement II: Frenkel defect leads to colour in ionic solids due to presence of F-centers.

Choose the **most appropriate** answer for the statements from the options given below:

- (A) **Statement I** is false, but **Statement II** is true
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Solution:

In an ionic compound showing Frenkel defect, the smaller ion (usually cation) is dislocated from its normal site to an interstitial site. It creates a vacancy defect at its original site and an interstitial defect at its new location.

F-centers occur due to metal excess defect. So, statement II is false.

Q49: The incorrect statement is:

- (A) Cl_2 is more reactive than ClF
- (B) F_2 is more reactive than ClF
- (C) On hydrolysis ClF forms HOCl and HF .
- (D) F_2 , is a stronger oxidizing agent than Cl_2 in aqueous solution

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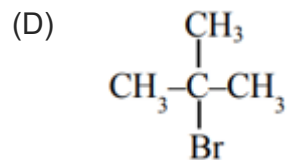
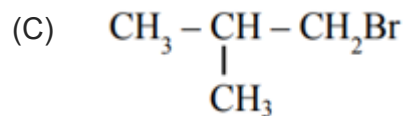
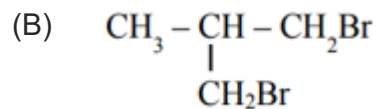
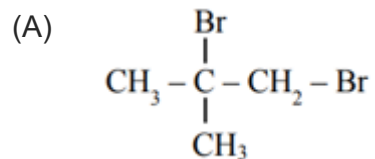
Solution:

(a) and (b): In general, interhalogen compounds are more reactive than halogens (except fluorine). This is because $\text{X}-\text{X}'$ bond in interhalogens is weaker than $\text{X}-\text{X}$ bond in halogens except $\text{F}-\text{F}$ bond. Thus, reactivity order: $\text{F}_2 > \text{ClF}$ (inter halogen) $> \text{Cl}_2$

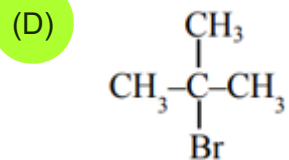
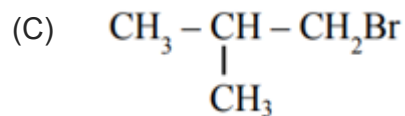
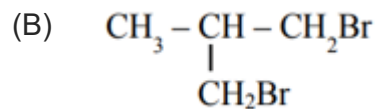
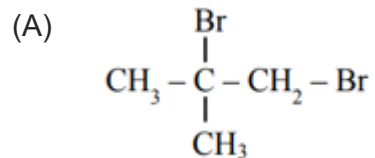
(c) $\text{ClF} + \text{H}_2\text{O} \rightarrow \text{HOCl} + \text{HF}$

(d) Oxidizing power in aqueous solution: $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$

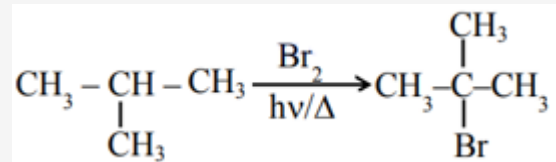
Q50: Excess of isobutane on reaction with Br_2 in presence of light at 125°C gives which one of the following as the major product?



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Solution:



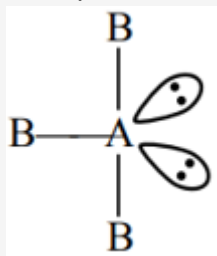
Q51: AB_3 , is an interhalogen T-shaped molecule. The number of lone pairs of electrons on A is _____. (Integer answer)

Q51: AB_3 , is an interhalogen T-shaped molecule. The number of lone pairs of electrons on A is _____. (Integer answer)

2.00

Solution:

T-shaped molecule means 3 sigma bond and 2 lone pairs of electrons on central atom.



Q52: These are physical properties of an element

- (a) Sublimation enthalpy
- (b) Ionisation enthalpy
- (c) Hydration enthalpy
- (d) Electron gain enthalpy

The total number of above properties that affect the reduction potential is _____ (Integer answer)

Q52: These are physical properties of an element

- (a) Sublimation enthalpy
- (b) Ionisation enthalpy
- (c) Hydration enthalpy
- (d) Electron gain enthalpy

The total number of above properties that affect the reduction potential is _____ (Integer answer)

Solution:

Sublimation enthalpy, ionisation enthalpy and hydration enthalpy affect the reduction potential.

Q53: Of the following four aqueous solutions, total number of those solutions whose freezing point is lower than that of 0.10 m $\text{C}_2\text{H}_5\text{OH}$ is _____ (Integer answer)

(i) 0.10 m $\text{Ba}_3(\text{PO}_4)_2$

(ii) 0.10 m Na_2SO_4

(iii) 0.10 m KCl

(iv) 0.10 m Li_3PO_4

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(i) 0.10 m $\text{Ba}_3(\text{PO}_4)_2$

(ii) 0.10 m Na_2SO_4

(iii) 0.10 m KCl

(iv) 0.10 m Li_3PO_4

4.00

Solution:

As 0.1 M $\text{C}_2\text{H}_5\text{OH}$ is non-dissociative ($i = 1$) and rest all salt given are dissociating electrolytes so in each case $i > 1$.

$$\Delta T_f = i \times K_f \times m$$

So, each will have lower freezing point.

Q54: The OH^- concentration in a mixture of 5.0 mL of 0.0504 M NH_4Cl and 2 mL of 0.0210 M NH_3 solution is $X \times 10^{-6}$ M. The value of x is _____. (Nearest integer)

[Given $K_w = 1 \times 10^{-14}$ and $K_b = 1.8 \times 10^{-5}$]

Q54: The OH^- concentration in a mixture of 5.0 mL of 0.0504 M NH_4Cl and 2 mL of 0.0210 M NH_3 solution is $X \times 10^{-6}$ M. The value of x is _____. (Nearest integer)

[Given $K_w = 1 \times 10^{-14}$ and $K_b = 1.8 \times 10^{-5}$]

3.00

Solution:

$$[\text{NH}_4^+] = 0.0504 \text{ \& } [\text{NH}_3] = 0.0210$$

$$\text{So } K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

$$[\text{OH}^-] = \frac{K_b \times [\text{NH}_3]}{[\text{NH}_4^+]} = 1.8 \times 10^{-5} \times \frac{2}{5} \times \frac{0.0210}{0.0504}$$

$$[\text{OH}^-] = 3 \times 10^{-6}$$

Q55: The number of 4f electrons in the ground state electronic configuration of Gd^{2+} is _____. [Atomic number of Gd = 64]

Q55: The number of 4f electrons in the ground state electronic configuration of Gd^{2+} is _____. [Atomic number of Gd = 64]

7.00

Solution:

The electronic configuration of

${}_{64}\text{Gd}$: $[\text{Xe}] 4f^7 5d^1 6s^2$

So, the electronic configuration of

${}_{64}\text{Gd}^{2+}$: $[\text{Xe}] 4f^7 5d^1 6s^0$

i.e., the number of 4f electrons in the ground state electronic configuration Gd^{2+} is 7.

Q56: The ratio of number of water molecules in Mohr's salt and potash alum is _____ $\times 10^{-1}$. (Integer answer)

Q56: The ratio of number of water molecules in Mohr's salt and potash alum is _____ $\times 10^{-1}$. (Integer answer)

5.00

Solution:

Mohr's salt: $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$

The number of water molecules in Mohr's salt = 6

Potash alum: $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$

The number of water molecules in potash alum = 12

So, the ratio of number of water molecules in Mohr's salt and potash alum = $6/12 = 1/2 = 0.5 = 5 \times 10^{-1}$

Q57: The following data was obtained for chemical reaction given below at 975 K.



	[NO] Mol L ⁻¹	[H ₂] Mol L ⁻¹	Rate Mol L ⁻¹ S ⁻¹
(a)	8×10^{-5}	8×10^{-5}	7×10^{-9}
(b)	24×10^{-5}	8×10^{-5}	2.1×10^{-8}
(c)	24×10^{-5}	32×10^{-5}	8.4×10^{-8}

The order of the reaction with respect to NO is _____ [Integer answer]

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	[NO] Mol L ⁻¹	[H ₂] Mol L ⁻¹	Rate Mol L ⁻¹ S ⁻¹
(a)	8×10^{-5}	8×10^{-5}	7×10^{-9}
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(c)	24×10^{-5}	32×10^{-5}	8.4×10^{-8}

The order of the reaction with respect to NO is _____ [Integer answer]

Solution:

In experiments (a) and (b) [H₂] is the same and [NO] is increased three times. The rate of the reaction is also increasing three times. Thus, the order with respect to NO is 1.

Q58: The Born-Haber cycle for KCl is evaluated with the following data:

$\Delta_f H^\ominus$ for KCl = $-436.7 \text{ kJ mol}^{-1}$;

$\Delta_{\text{sub}} H^\ominus$ for K = 89.2 kJ mol^{-1} ;

$\Delta_{\text{ionisation}} H^\ominus$ for K = $419.0 \text{ kJ mol}^{-1}$;

$\Delta_{\text{electron gain}} H^\ominus$ for $\text{Cl}_{(\text{g})}$ = $-348.6 \text{ kJ mol}^{-1}$;

$\Delta_{\text{bond}} H^\ominus$ for Cl_2 = $243.3 \text{ kJ mol}^{-1}$

The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is _____ (Nearest integer)

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$$\Delta_{\text{electron gain}} H^\ominus \text{ for Cl}_{(g)} = -348.6 \text{ kJ mol}^{-1};$$

$$\Delta_{\text{bond}} H^\ominus \text{ for Cl}_2 = 243.3 \text{ kJ mol}^{-1}$$

The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is _____ (Nearest integer)

718.00

Solution:

$$\Delta_f H^\ominus_{\text{KCl}} = \Delta_{\text{sub}} H^\ominus(\text{K}) + \Delta_{\text{ionization}} H^\ominus(\text{K}) + \frac{1}{2} \Delta_{\text{bond}} H^\ominus(\text{Cl}_2)$$

$$+ \Delta_{\text{electron gain}} H^\ominus(\text{Cl}) + \Delta_{\text{lattice}} H^\ominus(\text{KCl})$$

$$\Rightarrow -436.7 = 89.2 + 419.0 + \frac{1}{2}(243.3) + \{-348.6\} + \Delta_{\text{lattice}} H^\ominus(\text{KCl})$$

$$\Rightarrow \Delta_{\text{lattice}} H^\ominus(\text{KCl}) = -717.8 \text{ kJ mol}^{-1}$$

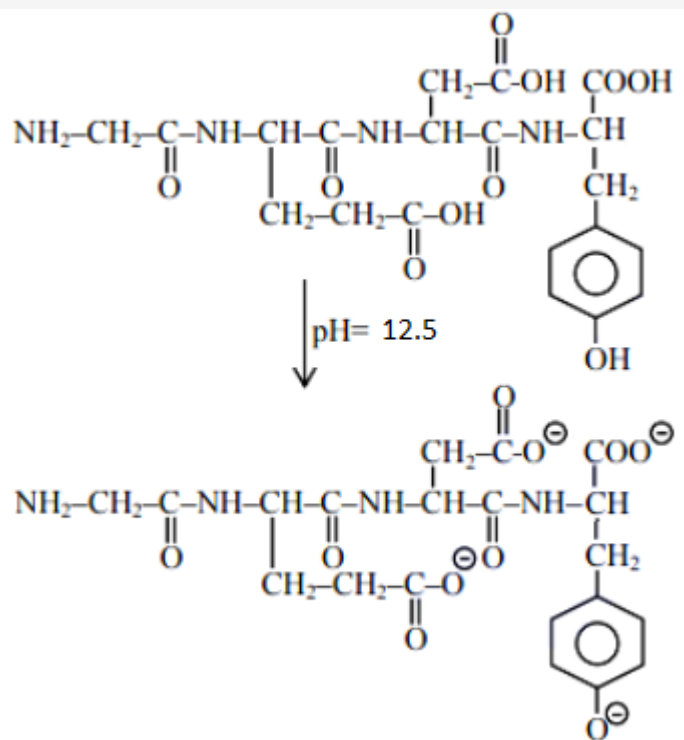
The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is 718 (Nearest integer).

Q59: The total number of negative charges in the tetrapeptide, Gly-Glu-Asp-Tyr, at pH 12.5 will be _____. (Integer answer)

Q59: The total number of negative charges in the tetrapeptide, Gly-Glu-Asp-Tyr, at pH 12.5 will be _____. (Integer answer)

4.00

Solution:



Total negative charge produced = 4.

Q60: An aqueous KCl solution of density 1.20 g mL^{-1} has a molality of 3.30 mol kg^{-1} . The molality of the solution in mol L^{-1} is _____ (Nearest integer)

[Molar mass of KCl = 74.5]

Q60: An aqueous KCl solution of density 1.20 g mL^{-1} has a molality of 3.30 mol kg^{-1} . The molarity of the solution in mol L^{-1} is _____ (Nearest integer)

[Molar mass of KCl = 74.5]

3.00

Solution:

1000 g solvent has 3.3 moles of KCl

1000 g solvent $\rightarrow 3.3 \times 74.5 \text{ g KCl} \rightarrow 245.85 \text{ g KCl}$

Weight of solution = 1245.85 g

Volume of solution = $\frac{1245.85}{1.2}$ ml

So, molarity = $\frac{3.3 \times 1.2}{1245.85} \times 1000 = 3.17$

Q61: If the sum of an infinite GP $a, ar, ar^2, ar^3, \dots$ is 15 and the sum of the squares of its each term is 150, then the sum of $ar^2, ar^4, ar^6, \dots$ is:

(A) $\frac{1}{2}$

(B) $\frac{5}{2}$

(C) $\frac{25}{2}$

(D) $\frac{9}{2}$

Q61: If the sum of an infinite GP $a, ar, ar^2, ar^3, \dots$ is 15 and the sum of the squares of its each term is 150, then the sum of $ar^2, ar^4, ar^6, \dots$ is:

(A) $\frac{1}{2}$

(B) $\frac{5}{2}$

(C) $\frac{25}{2}$

(D) $\frac{9}{2}$

Solution:

Sum of infinite terms:

$$\frac{a}{1-r} = 15 \quad \dots (i)$$

Series formed by square of terms:

$$a^2, a^2r^2, a^2r^4, a^2r^6, \dots$$

$$\text{Sum} = \frac{a^2}{1-r^2} = 150$$

$$\Rightarrow \frac{a}{1-r} \cdot \frac{a}{1+r} = 150$$

$$\Rightarrow 15 \cdot \frac{a}{1+r} = 150$$

$$\Rightarrow \frac{a}{1+r} = 10 \quad \dots (ii)$$

By (i) and (ii) $a = 12$; $r = \frac{1}{5}$

Now series: $ar^2, ar^4, ar^6, \dots$

$$\text{Sum} = \frac{ar^2}{1-r^2} = \frac{12 \cdot \left(\frac{1}{25}\right)}{1 - \frac{1}{25}} = \frac{1}{2}$$

Q62: Let ABC be a triangle with $A(-3, 1)$ and $\angle ACB = \theta$, $0 < \theta < \frac{\pi}{2}$. If the equation of the median through B is $2x + y - 3 = 0$ and the equation of angle bisector of C is $7x - 4y - 1 = 0$, then $\tan \theta$ is equal to :

(A) $\frac{4}{3}$

(B) $\frac{1}{2}$

(C) 2

(D) $\frac{3}{4}$

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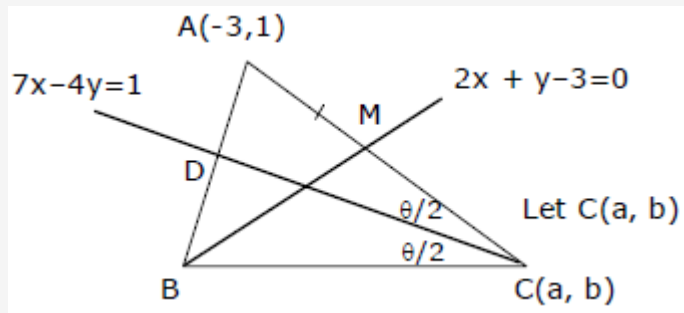
(A) $\frac{4}{3}$

(B) $\frac{1}{2}$

(C) 2

(D) $\frac{3}{4}$

Solution:



$$\therefore M\left(\frac{a-3}{2}, \frac{b+1}{2}\right) \text{ lies on } 2x + y - 3 = 0$$

$$\Rightarrow 2a + b = 11 \quad \dots (i)$$

$$\therefore C \text{ lies on } 7x - 4y = 1$$

$$\Rightarrow 7a - 4b = 1 \quad \dots (ii)$$

$$\therefore \text{by (i) and (ii): } a = 3, b = 5$$

$$\Rightarrow C(3, 5)$$

$$\therefore m_{AC} = \frac{2}{3}$$

$$\text{Also, } m_{CD} = \frac{7}{4}$$

$$\Rightarrow \tan \frac{\theta}{2} = \left| \frac{\frac{2}{3} - \frac{7}{4}}{1 + \frac{14}{12}} \right| \Rightarrow \tan \frac{\theta}{2} = \frac{1}{2}$$

$$\Rightarrow \tan \theta = \frac{2\left(\frac{1}{2}\right)}{1 - \frac{1}{4}} = \frac{4}{3}$$

Q63: Let $\theta \in \left(0, \frac{\pi}{2}\right)$. If the system of linear equations,

$$(1 + \cos^2 \theta) x + (\sin^2 \theta) y + (4 \sin 3\theta) z = 0$$

$$(\cos^2 \theta) x + (1 + \sin^2 \theta) y + (4 \sin 3\theta) z = 0$$

$$(\cos^2 \theta) x + (\sin^2 \theta) y + (1 + 4 \sin 3\theta) z = 0$$

Has a non-trivial solution, then the value of θ is:

(A) $\frac{4\pi}{9}$

(B) $\frac{\pi}{18}$

(C) $\frac{5\pi}{18}$

(D) $\frac{7\pi}{18}$

Q63: Let $\theta \in \left(0, \frac{\pi}{2}\right)$. If the system of linear equations,

$$(1 + \cos^2 \theta) x + (\sin^2 \theta) y + (4 \sin 3\theta) z = 0$$

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Has a non-trivial solution, then the value of θ is:

(A) $\frac{4\pi}{9}$

(B) $\frac{\pi}{18}$

(C) $\frac{5\pi}{18}$

(D) $\frac{7\pi}{18}$

Solution:

Case -I

$$\begin{vmatrix} 1 + \cos^2 \theta & \sin^2 \theta & 4 \sin 3\theta \\ \cos^2 \theta & 1 + \sin^2 \theta & 4 \sin 3\theta \\ \cos^2 \theta & \sin^2 \theta & 1 + 4 \sin 3\theta \end{vmatrix} = 0$$

$$C_1 \rightarrow C_1 + C_2$$

$$\begin{vmatrix} 2 & \sin^2 \theta & 4 \sin 3\theta \\ 2 & 1 + \sin^2 \theta & 4 \sin 3\theta \\ 1 & \sin^2 \theta & 1 + 4 \sin 3\theta \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$$

$$\begin{vmatrix} 0 & -1 & 0 \\ 1 & 1 & -1 \\ 1 & \sin^2 \theta & 1 + 4 \sin^3 \theta \end{vmatrix} = 0$$

$$\text{Or } 4 \sin 3\theta = -2$$

$$\sin 3\theta = -\frac{1}{2}$$

$$\theta = \frac{7\pi}{18}$$

Q64: The sum of solutions of the equation

$$\frac{\cos x}{1+\sin x} = |\tan 2x|, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}, -\frac{\pi}{4}\right\} \text{ is:}$$

(A) $-\frac{11\pi}{30}$

(B) $-\frac{7\pi}{30}$

(C) $-\frac{\pi}{15}$

(D) $\frac{\pi}{10}$

Q64: The sum of solutions of the equation

$$\frac{\cos x}{1+\sin x} = |\tan 2x|, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}, -\frac{\pi}{4}\right\} \text{ is:}$$

(A) $-\frac{11\pi}{30}$

(B) $-\frac{7\pi}{30}$

(C) $-\frac{\pi}{15}$

(D) $\frac{\pi}{10}$

Solution:

$$\frac{\cos x}{1+\sin x} = |\tan 2x|$$

$$\Rightarrow \frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\left(\cos \frac{x}{2} + \sin \frac{x}{2}\right)^2} = |\tan 2x|$$

$$\Rightarrow \tan^2 \left(\frac{\pi}{4} - \frac{x}{2}\right) = \tan^2 2x$$

$$\Rightarrow 2x = n\pi \pm \left(\frac{\pi}{4} - \frac{x}{2}\right)$$

$$\Rightarrow x = \frac{-3\pi}{10}, \frac{-\pi}{6}, \frac{\pi}{10}$$

$$\text{Or sum} = -\frac{11\pi}{30}$$

Q65: Let A and B be independent events such that $P(A) = p$, $P(B) = 2p$. The largest value of p, for which $P(\text{exactly one of A, B occurs}) = \frac{5}{9}$, is:

(A) $\frac{4}{9}$

(B) $\frac{2}{9}$

(C) $\frac{1}{3}$

(D) $\frac{5}{12}$

Q65: Let A and B be independent events such that $P(A) = p$, $P(B) = 2p$. The largest value of p, for which $P(\text{exactly one of A, B occurs}) = \frac{5}{9}$, is:

(A) $\frac{4}{9}$

(B) $\frac{2}{9}$

(C) $\frac{1}{3}$

(D) $\frac{5}{12}$

Solution:

$$\begin{aligned} &P(\text{Exactly one of A or B}) \\ &= P(A \cap \bar{B}) + P(\bar{A} \cap B) = \frac{5}{9} \\ &= P(A)P(\bar{B}) + P(\bar{A})P(B) = \frac{5}{9} \\ &\Rightarrow P(A)(1 - P(B)) + (1 - P(A))P(B) = \frac{5}{9} \\ &\Rightarrow p(1 - 2p) + (1 - p)2p = \frac{5}{9} \\ &\Rightarrow 36p^2 - 27p + 5 = 0 \\ &\Rightarrow p = \frac{1}{3} \text{ or } \frac{5}{12} \\ &P_{\max} = \frac{5}{12} \end{aligned}$$

Q66: If the truth value of the Boolean expression $((p \vee q) \wedge (q \rightarrow r) \wedge (\sim r)) \rightarrow (p \wedge q)$ is false, then the truth values of the statements p , q , r respectively can be:

- (A) FFT
- (B) FTF
- (C) TFT
- (D) TFF

Q66: If the truth value of the Boolean expression $((p \vee q) \wedge (q \rightarrow r) \wedge (\sim r)) \rightarrow (p \wedge q)$ is false, then the truth values of the statements p , q , r respectively can be:

(A) FFT

(B) FTF

(C) TFT

(D) TFF

Solution:

p	q	r	$\underbrace{p \vee q}_a$	$\underbrace{q \rightarrow r}_b$	$a \wedge b$	$\sim r$	$\underbrace{a \wedge b \wedge (\sim r)}_c$	$\underbrace{p \wedge q}_d$	$c \rightarrow d$
T	F	T	T	T	T	F	F	F	T
F	F	T	F	T	F	F	F	F	T
T	F	F	T	T	T	T	T	F	F
F	T	F	T	F	F	T	F	F	T

Q67: The value of $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2}$ is:

(A) $\frac{1}{4} \tan^{-1}(4)$

(B) $\tan^{-1}(4)$

(C) $\frac{1}{2} \tan^{-1}(2)$

(D) $\frac{1}{2} \tan^{-1}(4)$

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(A) $\frac{1}{4} \tan^{-1}(4)$

(B) $\tan^{-1}(4)$

(C) $\frac{1}{2} \tan^{-1}(2)$

(D) $\frac{1}{2} \tan^{-1}(4)$

Solution:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{1}{1 + 4\left(\frac{r}{n}\right)^2}$$

$$\Rightarrow L = \int_0^2 \frac{1}{1 + 4x^2} dx$$

$$\Rightarrow L = \frac{1}{2} \tan^{-1} 4$$

Q68: Let $y = y(x)$ be a solution curve of the differential equation

$(y + 1) \tan^2 x \, dx + \tan x \, dy + y \, dx = 0$, $x \in (0, \frac{\pi}{2})$. If $\lim_{x \rightarrow 0^+} x y(x) = 1$, then the

value of $y\left(\frac{\pi}{4}\right)$ is:

(A) $\frac{\pi}{4} - 1$

(B) $\frac{\pi}{4} + 1$

(C) $\frac{\pi}{4}$

(D) $-\frac{\pi}{4}$

Q68: Let $y = y(x)$ be a solution curve of the differential equation

$(y + 1) \tan^2 x \, dx + \tan x \, dy + y \, dx = 0$, $x \in (0, \frac{\pi}{2})$. If $\lim_{x \rightarrow 0^+} x y(x) = 1$, then the

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(A) $\frac{\pi}{4} - 1$

(B) $\frac{\pi}{4} + 1$

(C) $\frac{\pi}{4}$

(D) $-\frac{\pi}{4}$

Solution:

$$(y + 1) \tan^2 x \, dx + \tan x \, dy + y \, dx = 0$$

$$\text{Or } \frac{dy}{dx} + \frac{\sec^2 x}{\tan x} \cdot y = -\tan x$$

$$IF = e^{\int \frac{\sec^2 x}{\tan x} dx} = e^{\ln \tan x} = \tan x$$

$$\therefore y \tan x = -\int \tan^2 x \, dx$$

$$\text{Or } y \tan x = -\tan x + x + C$$

$$\text{Or } y = -1 + \frac{x}{\tan x} + \frac{C}{\tan x}$$

$$\text{Or } \lim_{x \rightarrow 0} xy = -x + \frac{x^2}{\tan x} + \frac{Cx}{\tan x} = 1$$

$$\text{Or } C = 1$$

$$y(x) = \cot x + x \cot x - 1$$

$$y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$$

Q69: The mean and standard deviation of 20 observations were calculated as 10 and 2.5 respectively. It was found that by mistake one data value was taken as 25 instead of 35. If α and $\sqrt{\beta}$ are the mean and standard deviation respectively for correct data, then (α, β) is:

- (A) (11, 25)
- (B) (11, 26)
- (C) (10.5, 26)
- (D) (10.5, 25)

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- (A) (11, 25)
- (B) (11, 26)
- (C) (10.5, 26)
- (D) (10.5, 25)

Solution:

Given:

$$\text{Mean } (\bar{x}) = \frac{\sum x_i}{20} = 10$$

$$\text{Or } \sum x_i = 200 \text{ (incorrect)}$$

$$\text{Or } 200 - 25 + 35 = \sum x_i \text{ (correct)}$$

$$\text{Now correct } \bar{x} = \frac{210}{20} = 10.5$$

$$\text{Again given S.D} = 2.5 (\sigma)$$

$$\sigma^2 = \frac{\sum x_i^2}{20} - (10)^2 = (2.5)^2$$

$$\text{Or } \sum x_i^2 = 2125 - 25^2 + 35^2$$

$$= 2725 \text{ (correct)}$$

$$\therefore \text{Correct } \sigma^2 = \frac{2725}{20} - (10.5)^2$$

$$\sigma^2 = 26$$

$$\text{Or } \sigma = \sqrt{26}$$

$$\therefore \alpha = 10.5, \beta = 26$$

Q70: Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$, $0 < x < 1$. Then:

(A) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$

(B) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$

(C) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$

(D) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$

Q70: Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$, $0 < x < 1$. Then:

(A) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$

(B) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$

(C) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$

(D) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$

Solution:

$$f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$$

$$\cot^{-1}\sqrt{\frac{1-x}{x}} = \sin^{-1}\sqrt{x}$$

$$\text{Or } f(x) = \cos(2\tan^{-1}\sqrt{x})$$

$$= \cos \tan^{-1}\left(\frac{2\sqrt{x}}{1-x}\right)$$

$$f(x) = \frac{1-x}{1+x}$$

$$\text{Now, } f'(x) = \frac{-2}{(1+x)^2}$$

$$\text{Or } f'(x)(1-x)^2 = -2\left(\frac{1-x}{1+x}\right)^2$$

$$\text{Or } (1-x)^2 f'(x) + 2(f(x))^2 = 0$$

Q71: The sum of the series $\frac{1}{x+1} + \frac{2}{x^2+1} + \frac{2^2}{x^4+1} + \dots + \frac{2^{100}}{x^{2^{100}}+1}$ when $x = 2$ is:

(A) $1 - \frac{2^{101}}{4^{101}-1}$

(B) $1 + \frac{2^{101}}{4^{101}-1}$

(C) $1 - \frac{2^{100}}{4^{100}-1}$

(D) $1 + \frac{2^{100}}{4^{100}-1}$

Q71: The sum of the series $\frac{1}{x+1} + \frac{2}{x^2+1} + \frac{2^2}{x^4+1} + \dots + \frac{2^{100}}{x^{2^{100}}+1}$ when $x = 2$ is:

(A) $1 - \frac{2^{101}}{4^{101}-1}$

(B) $1 + \frac{2^{101}}{4^{101}-1}$

(C) $1 - \frac{2^{100}}{4^{100}-1}$

(D) $1 + \frac{2^{100}}{4^{100}-1}$

Solution:

$$\begin{aligned} S &= \frac{1}{x+1} + \frac{2}{x^2+1} + \frac{2^2}{x^4+1} + \dots + \frac{2^{100}}{x^{2^{100}}+1} \\ S + \frac{1}{1-x} &= \frac{1}{1-x} + \frac{1}{x+1} + \dots = \frac{2}{1-x^2} + \frac{2}{1+x^2} + \dots \\ S + \frac{1}{1-x} &= \frac{2^{101}}{1-x^{2^{101}}} \end{aligned}$$

Put $x = 2$

$$S = 1 - \frac{2^{101}}{4^{101}-1}$$

Q72: If ${}^{20}C_r$ is the co-efficient of x^r in the expansion of $(1 + x)^{20}$, then the value of

$\sum_{r=0}^{20} r^2 {}^{20}C_r$ is equal to :

- (A) 420×2^{19}
- (B) 420×2^{18}
- (C) 380×2^{18}
- (D) 380×2^{19}

Q72: If ${}^{20}C_r$ is the co-efficient of x^r in the expansion of $(1+x)^{20}$, then the value of

$\sum_{r=0}^{20} r^2 {}^{20}C_r$ is equal to :

(A) 420×2^{19}

(B) 420×2^{18}

(C) 380×2^{18}

(D) 380×2^{19}

Solution:

$$\begin{aligned} & \sum_{r=0}^{20} r^2 \cdot {}^{20}C_r \\ & \sum (r(r-1) + r) \cdot {}^{20}C_r \\ & \sum r(r-1) \cdot \frac{20 \times 19}{r(r-1)} \cdot {}^{18}C_r + r \cdot \frac{20}{r} \sum {}^{19}C_{r-1} \\ & \Rightarrow 20 \times 19 \cdot 2^{18} + 20 \cdot 2^{19} \\ & \Rightarrow 420 \times 2^{18} \end{aligned}$$

Q73: A plane P contains the line $x + 2y + 3z + 1 = 0 = x - y - z - 6$, and is perpendicular to the plane $-2x + y + z + 8 = 0$. Then which of the following points lies on P?

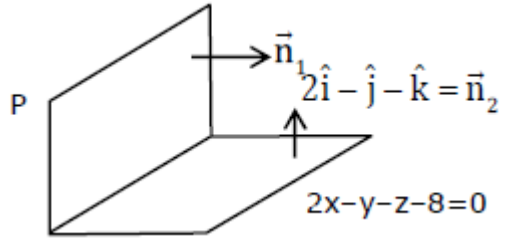
- (A) (1, 0, 1)
- (B) (2, -1, 1)
- (C) (-1, 1, 2)
- (D) (0, 1, 1)

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- (A) (1, 0, 1)
- (B) (2, -1, 1)
- (C) (-1, 1, 2)
- (D) (0, 1, 1)

Solution:

Equation of plane P can be assumed as



$$P : x + 2y + 3z + 1 + \lambda(x - y - z - 6) = 0$$

$$\Rightarrow P : (1 + \lambda)x + (2 - \lambda)y + (3 - \lambda)z + 1 - 6\lambda = 0$$

$$\vec{n}_1 = (1 + \lambda)\hat{i} + (2 - \lambda)\hat{j} + (3 - \lambda)\hat{k}$$

$$\therefore \vec{n}_1 \cdot \vec{n}_2 = 0$$

$$\Rightarrow 2(1 + \lambda) - (2 - \lambda) - (3 - \lambda) = 0$$

$$\Rightarrow 2 + 2\lambda - 2 + \lambda - 3 + \lambda = 0$$

$$\Rightarrow \lambda = \frac{3}{4}$$

$$\Rightarrow P : \frac{7x}{4} + \frac{5y}{4} + \frac{9z}{4} - \frac{14}{4} = 0$$

$$\Rightarrow 7x + 5y + 9z = 14$$

$(0, 1, 1)$ lies on P

Q74: On the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$ let P be a point in the second quadrant such that the tangent at P to the ellipse is perpendicular to the line $x + 2y = 0$. Let S and S' be the foci of the ellipse and e be its eccentricity. If A is the area of the triangle SPS' then, the value of $(5 - e^2) \cdot A$ is:

- (A) 24
- (B) 6
- (C) 14
- (D) 12

Q74: On the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$ let P be a point in the second quadrant such that the tangent at P to the ellipse is perpendicular to the line $x + 2y = 0$. Let S and S' be the foci of the ellipse and e be its eccentricity. If A is the area of the triangle SPS' then, the value of $(5 - e^2) \cdot A$ is:

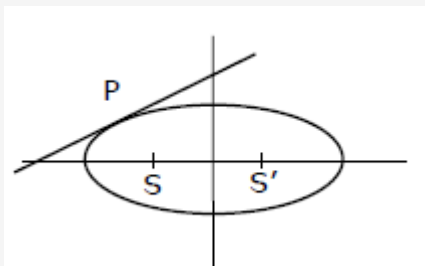
(A) 24

(B) 6

(C) 14

(D) 12

Solution:



Equation of tangent : $y = 2x + 6$ at P

$$\therefore P\left(-\frac{8}{3}, \frac{2}{3}\right)$$

$$e = \frac{1}{\sqrt{2}}$$

S and $S' = (-2, 0)$ and $(2, 0)$

Area of $\Delta SPS' = \frac{1}{2} \times 4 \times \frac{2}{3}$

$$A = \frac{4}{3}$$

$$\therefore (5 - e^2) A = \left(5 - \frac{1}{2}\right) \frac{4}{3} = 6$$

Q75: The value of $\int_{\frac{-1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left(\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right)^{\frac{1}{2}} dx$ is:

(A) $\log_e 4$

(B) $\log_e 16$

(C) $4\log_e (3 + 2\sqrt{2})$

(D) $2\log_e 16$

Q75: The value of $\int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left(\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right)^{\frac{1}{2}} dx$ is:

- (A) $\log_e 4$
- (B) $\log_e 16$
- (C) $4\log_e (3 + 2\sqrt{2})$
- (D) $2\log_e 16$

Solution:

$$I = \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left(\left(\frac{x+1}{x-1} - \frac{x-1}{x+1} \right)^2 \right)^{\frac{1}{2}} dx$$

$$I = \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left| \frac{4x}{x^2-1} \right| dx \Rightarrow I = 2.4 \int_0^{\frac{1}{\sqrt{2}}} \left| \frac{x}{x^2-1} \right| dx$$

$$\Rightarrow I = -4 \int_0^{\frac{1}{\sqrt{2}}} \frac{2x}{x^2-1} dx \Rightarrow I = -4 \ln |x^2 - 1|_0^{\frac{1}{\sqrt{2}}}$$

$$\Rightarrow I = 4 \ln 2 \Rightarrow I = \ln 16$$

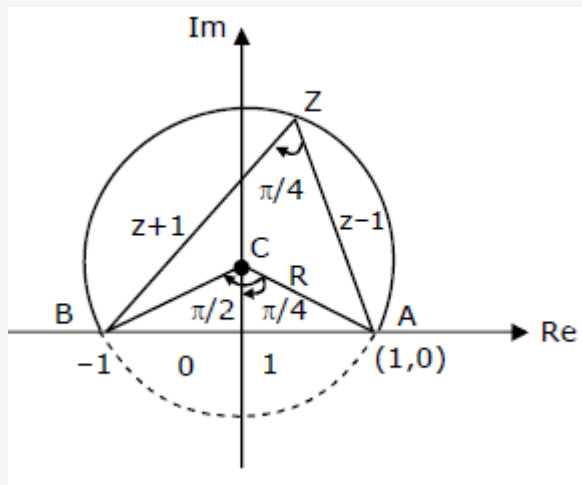
Q76: The equation $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circle with :

- (A) centre at $(0, -1)$ and radius $\sqrt{2}$
- (B) centre at $(0, 1)$ and radius 2
- (C) centre at $(0, 1)$ and radius $\sqrt{2}$
- (D) centre at $(0, 0)$ and radius $\sqrt{2}$

Q76: The equation $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circle with :

- (A) centre at $(0, -1)$ and radius $\sqrt{2}$
- (B) centre at $(0, 1)$ and radius 2
- (C) centre at $(0, 1)$ and radius $\sqrt{2}$
- (D) centre at $(0, 0)$ and radius $\sqrt{2}$

Solution:



In ΔOAC

$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{AC}$$

$$\Rightarrow AC = \sqrt{2}$$

$$\text{Also, } \tan \frac{\pi}{4} = \frac{OA}{OC} = \frac{1}{OC}$$

$$\Rightarrow OC = 1$$

$$\therefore \text{Centre } (0, 1), \text{ Radius} = \sqrt{2}$$

Q77: If a line along a chord of the circle $4x^2 + 4y^2 + 120x + 675 = 0$, passes through the point $(-30, 0)$ and is tangent to the parabola $y^2 = 30x$, then the length of this chord is:

- (A) 5
- (B) $3\sqrt{5}$
- (C) 7
- (D) $5\sqrt{3}$

Q77: If a line along a chord of the circle $4x^2 + 4y^2 + 120x + 675 = 0$, passes through the point $(-30, 0)$ and is tangent to the parabola $y^2 = 30x$, then the length of this chord is:

(A) 5

(B) $3\sqrt{5}$

(C) 7

(D) $5\sqrt{3}$

Solution:

Equation of tangent to $y^2 = 30x$

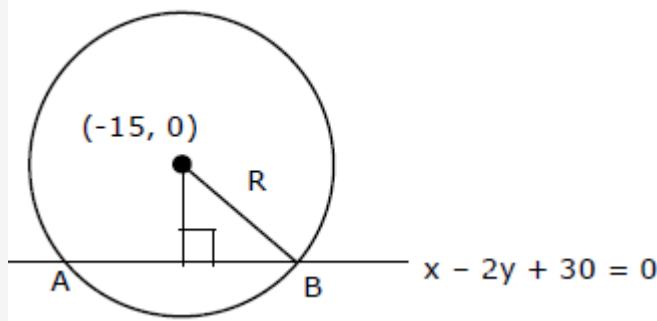
$$y = mx + \frac{30}{4m}$$

$$\text{Pass through } (-30, 0) : 0 = -30m + \frac{30}{4m} \Rightarrow m^2 = \frac{1}{4}$$

$$\Rightarrow m = \frac{1}{2} \text{ or } m = -\frac{1}{2}$$

$$\text{At } m = \frac{1}{2} : y = \frac{x}{2} + 15$$

$$\Rightarrow x - 2y + 30 = 0$$



$$P = \frac{15}{\sqrt{5}}$$

$$l_{AB} = 2\sqrt{R^2 - P^2} = 2\sqrt{\frac{225}{4} - \frac{225}{5}}$$

$$\Rightarrow l_{AB} = 30\sqrt{\frac{1}{20}} = \frac{15}{\sqrt{5}} = 3\sqrt{5}$$

Q78: Let $\bar{a} = \hat{i} + \hat{j} + \hat{k}$ and $\bar{b} = \hat{j} - \hat{k}$. If $\bar{c}$ is a vector such that $\bar{a} \times \bar{c} = \bar{b}$ and $\bar{a} \cdot \bar{c} = 3$, then $\bar{a} \cdot (\bar{b} \times \bar{c})$ is equal to :

(A) -2

(B) 6

(C) 2

(D) -6

Q78: Let $\bar{a} = \hat{i} + \hat{j} + \hat{k}$ and $\bar{b} = \hat{j} - \hat{k}$. If $\bar{c}$ is a vector such that $\bar{a} \times \bar{c} = \bar{b}$ and $\bar{a} \cdot \bar{c} = 3$, then $\bar{a} \cdot (\bar{b} \times \bar{c})$ is equal to :

(A) -2

(B) 6

(C) 2

(D) -6

Solution:

$$|\bar{a}| = \sqrt{3}; \bar{a} \cdot \bar{c} = 3; \bar{a} \times \bar{b} = -2\hat{i} + \hat{j} + \hat{k}, \bar{a} \times \bar{c} = \bar{b}$$

Cross with $\bar{a}$.

$$\bar{a} \cdot (\bar{a} \times \bar{c}) = \bar{a} \times \bar{b}$$

$$\Rightarrow (\bar{a} \cdot \bar{c}) \bar{a} - a^2 \bar{c} = \bar{a} \times \bar{b}$$

$$\Rightarrow 3\bar{a} - 3\bar{c} = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow 3\hat{i} + 3\hat{j} + 3\hat{k} - 3\bar{c} = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow \bar{c} = \frac{5\hat{i}}{3} + \frac{2\hat{j}}{3} + \frac{2\hat{k}}{3}$$

$$\therefore \bar{a} \cdot (\bar{b} \times \bar{c}) = (\bar{a} \times \bar{b}) \cdot \bar{c} = \frac{-10}{3} + \frac{2}{3} + \frac{2}{3} = -2$$

Q79: If $A = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$, $i = \sqrt{-1}$ and $Q = A^T B A$, then the inverse of the matrix $A Q^{2021} A^T$ is equal to :

(A) $\begin{pmatrix} \frac{1}{\sqrt{5}} & -2021 \\ 2021 & \frac{1}{\sqrt{5}} \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & -2021i \\ 0 & 1 \end{pmatrix}$

Q79: If $A = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$, $i = \sqrt{-1}$ and $Q = A^T B A$, then the inverse of the matrix $A Q^{2021} A^T$ is equal to :

(A) $\begin{pmatrix} \frac{1}{\sqrt{5}} & -2021 \\ 2021 & \frac{1}{\sqrt{5}} \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & -2021i \\ 0 & 1 \end{pmatrix}$

Solution:

$$AA^T = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$$

$$AA^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$Q^2 = A^T B A A^T B A = A^T B I B A$$

$$\Rightarrow Q^2 = A^T B^2 A$$

$$Q^3 = A^T B^2 A A^T B A$$

$$\Rightarrow Q^3 = A^T B^3 A$$

$$\text{Similarly, } Q^{2021} = A^T B^{2021} A \quad \dots (1)$$

$$\text{Now, } B^2 = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2i & 1 \end{pmatrix}$$

$$B^3 = \begin{pmatrix} 1 & 0 \\ 2i & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix} \Rightarrow B^3 = \begin{pmatrix} 1 & 0 \\ 3i & 1 \end{pmatrix}$$

$$\text{Similarly } B^{2021} = \begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$$

$$\therefore A Q^{2021} A^T = A A^T B^{2021} A A^T = I B^{2021} I$$

$$\Rightarrow A Q^{2021} A^T = B^{2021} = \begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$$

$$\therefore (A Q^{2021} A^T)^{-1} = \begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$$

Q80: Out of all the patients in a hospital **89%** are found to be suffering from heart ailment and **98%** are suffering from lungs infection. If ***K*%** of them are suffering from both ailments, then **K** cannot belong to the set:

- (A) {80, 83, 86, 89}
- (B) {79, 81, 83, 85}
- (C) {84, 87, 90, 93}
- (D) {84, 86, 88, 90}

Q80: Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If $K\%$ of them are suffering from both ailments, then K cannot belong to the set:

(A) {80, 83, 86, 89}

(B) {79, 81, 83, 85}

(C) {84, 87, 90, 93}

(D) {84, 86, 88, 90}

Solution:

$$n(A \cup B) \geq n(A) + n(B) - n(A \cap B)$$

$$100 \geq 89 + 98 - n(A \cup B)$$

$$n(A \cup B) \geq 87$$

$$87 \leq n(A \cup B) \leq 89$$

Q81: The sum of all integral values of k ($k \neq 0$) for which the equation $\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$ in x has no real roots, is _____.

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66.00

Solution:

$$\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$$

$$x \in \mathbb{R} - \{1, 2\}$$

$$\Rightarrow k(2x - 4 - x + 1) = 2(x^2 - 3x + 2)$$

$$\Rightarrow k(x - 3) = 2(x^2 - 3x + 2)$$

$$\text{For } x \neq 3, k = 2\left(x - 3 + \frac{2}{x-3} + 3\right)$$

$$x - 3 + \frac{2}{x-3} \geq 2\sqrt{2} \quad \forall x > 3$$

$$\text{And } x - 3 + \frac{2}{x-3} \leq -2\sqrt{2} \quad \forall x < 3$$

$$\Rightarrow 2\left(x - 3 + \frac{2}{x-3} + 3\right) \in (-\infty, 6 - 4\sqrt{2}] \cup [6 + 4\sqrt{2}, \infty)$$

For no real roots

$$k \in (6 - 4\sqrt{2}, 6 + 4\sqrt{2}) - \{0\}$$

$$\text{Integral } k \in \{1, 2, \dots, 11\}$$

$$\text{Sum of } k = 66$$

Q82: The locus of a point, which moves such that the sum of squares of its distances from the points $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$ is 18 units, is a circle of diameter d . Then d^2 is equal to _____.

Q82: The locus of a point, which moves such that the sum of squares of its distances from the points $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$ is 18 units, is a circle of diameter d . Then d^2 is equal to _____.

16.00

Solution:

Let $P(x, y)$

$$x^2 + y^2 + x^2 + (y - 1)^2 + (x - 1)^2 + y^2 + (x - 1)^2 + (y - 1)^2 = 18$$

$$\Rightarrow 4(x^2 + y^2) - 4y - 4x = 14$$

$$\Rightarrow x^2 + y^2 - x - y - \frac{7}{2} = 0$$

$$d = 2\sqrt{\frac{1}{4} + \frac{1}{4} + \frac{7}{2}}$$

$$\Rightarrow d^2 = 16$$

Q83: If $y = y(x)$ is an implicit function of x such that $\log_e (x + y) = 4xy$, then $\frac{d^2y}{dx^2}$ at $x = 0$ is equal to _____.

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40.00

Solution:

$$\ln(x + y) = 4xy \text{ (At } x = 0, y = 1)$$

$$x + y = e^{4xy}$$

$$\Rightarrow 1 + \frac{dy}{dx} = e^{4xy} \left(4x \frac{dy}{dx} + 4y \right)$$

$$\text{At } x = 0, \frac{dy}{dx} = 3$$

$$\frac{d^2y}{dx^2} = e^{4xy} \left(4x \frac{dy}{dx} + 4y \right)^2 + e^{4xy} \left(4x \frac{d^2y}{dx^2} + 8 \frac{dy}{dx} \right)$$

$$\text{At } x = 0, \frac{d^2y}{dx^2} = e^0 (4)^2 + e^0 (24)$$

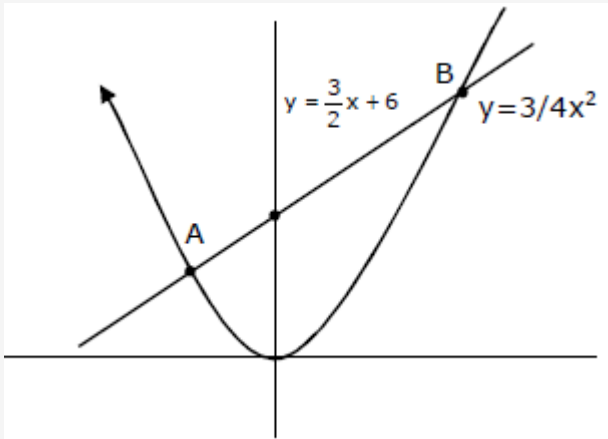
$$\Rightarrow \frac{d^2y}{dx^2} = 40$$

Q84: The area of the region $S = \{(x, y) : 3x^2 \leq 4y \leq 6x + 24\}$ is _____.

Q84: The area of the region $S = \{(x, y) : 3x^2 \leq 4y \leq 6x + 24\}$ is _____.

27.00

Solution:



For A and B

$$3x^2 = 6x + 24$$

$$\Rightarrow x^2 - 2x - 8 = 0$$

$$\Rightarrow x = -2, 4$$

$$\text{Area} = \int_{-2}^4 \left(\frac{3}{2}x + 6 - \frac{3}{4}x^2 \right) dx$$

$$= \left[\frac{3x^2}{4} + 6x - \frac{x^3}{4} \right]_{-2}^4$$

$$= 27$$

Q85: Let $a, b \in \mathbb{R}$, $b \neq 0$. Define a function

$$f(x) = \begin{cases} a \sin \frac{\pi}{2}(x-1) & \text{for } x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3} & \text{for } x > 0 \end{cases}$$

If f is continuous at $x = 0$, then $10 - ab$ is equal to _____.

Q85: Let $a, b \in \mathbb{R}$, $b \neq 0$. Define a function

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If f is continuous at $x = 0$, then $10 - ab$ is equal to _____.

14.00

Solution:

$$f(x) = \begin{cases} a \sin \frac{\pi}{2}(x-1) & \text{for } x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3} & \text{for } x > 0 \end{cases}$$

For continuity at '0'

$$\lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\tan 2x - \sin 2x}{bx^3} = -a$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\frac{8x^3}{3} + \frac{8x^3}{3!}}{bx^3} = -a$$

$$\Rightarrow 8 \left(\frac{1}{3} + \frac{1}{3!} \right) = -ab$$

$$\Rightarrow 4 = -ab$$

$$\Rightarrow 10 - ab = 14$$

Q86: If ${}^1P_1 + 2.{}^2P_2 + 3.{}^3P_3 + \dots\dots\dots 15.{}^{15}P_{15} = {}^ap_r - s$, $0 \leq s \leq 1$, then ${}^{q+s}C_{r-s}$ is equal to _____.

Q86: If ${}^1P_1 + 2.{}^2P_2 + 3.{}^3P_3 + \dots + 15.{}^{15}P_{15} = {}^qp_r - s$, $0 \leq s \leq 1$, then ${}^{q+s}C_{r-s}$ is equal to _____.

136.00

Solution:

$$\begin{aligned}
 & {}^1P_1 + 2.{}^2P_2 + 3.{}^3P_3 + \dots + 15.{}^{15}P_{15} \\
 &= 1! + 2.2! + 3.3! + \dots + 15 \times 15! \\
 &= \sum_{r=1}^{15} (r+1-1)r! \\
 &= \sum_{r=1}^{15} (r+1)! - (r)! \\
 &= 16! - 1 \\
 &= {}^{16}P_{16} - 1 \\
 &\Rightarrow q = r = 16, s = 1 \\
 &{}^{q+s}C_{r-s} = {}^{17}C_{15} = 136
 \end{aligned}$$

Q87: A wire of length 36 m is cut into two pieces, one of the pieces is bent to form a square and the other is bent to form a circle. If the sum of the areas of the two figures is minimum, and the circumference of the circle is K (meter), then $\left(\frac{4}{\pi} + 1\right) k$ is equal to _____.

Q87: A wire of length 36 m is cut into two pieces, one of the pieces is bent to form a square and the other is bent to form a circle. If the sum of the areas of the two figures is minimum, and the circumference of the circle is K (meter), then $\left(\frac{4}{\pi} + 1\right) k$ is equal to _____.

36.00

Solution:

$$\text{Let } x + y = 36$$

x is perimeter of square and y is perimeter of circle side of square = $\frac{x}{4}$

$$\text{Radius of circle} = \frac{y}{2\pi}$$

$$\text{Sum areas} = \left(\frac{x}{4}\right)^2 + \pi\left(\frac{y}{2\pi}\right)^2$$

$$= \frac{x^2}{16} + \frac{(36-x)^2}{4\pi}$$

$$\text{For min area: } x = \frac{144}{\pi+4}$$

$$\Rightarrow \text{Radius} = y = 36 - \frac{144}{\pi+4}$$

$$\Rightarrow k = \frac{36\pi}{\pi+4}$$

$$\left(\frac{4}{\pi} + 1\right) k = 36$$

Q88: Let the line L be the projection of the line :

$$\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$$

In the plane $x - 2y - z = 3$. If d is the distance of the point $(0, 0, 6)$ from L, then d^2 is equal to _____.

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Solution:

$$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$$

For foot of $\perp$ r of $(1, 3, 4)$ on $x - 2y - z - 3 = 0$

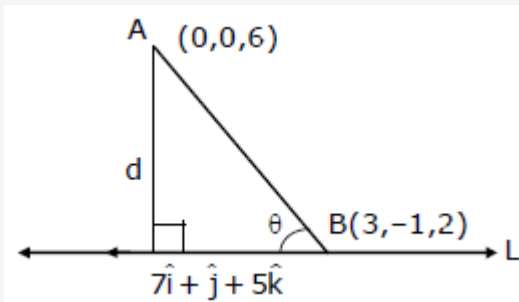
$$(1+t) - 2(3-2t) - (4-t) - 3 = 0$$

$$\Rightarrow t = 2$$

So foot of $\perp = (3, -1, 2)$

And point of intersection of L_1 with plane is $(-11, -3, -8)$

dr's of L is $\langle 14, 2, 10 \rangle \cong \langle 7, 1, 5 \rangle$



$$d = AB \sin \theta = \frac{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -4 \\ 7 & 1 & 5 \end{vmatrix}}{\sqrt{7^2+1^2+5^2}}$$

$$\Rightarrow d = \frac{1^2+(43)^2+(10)^2}{49+1+25} = 26$$

Q89: Let $z = \frac{1-i\sqrt{3}}{2}$, $i = \sqrt{-1}$. Then the value of

$21 + \left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \left(z^3 + \frac{1}{z^3}\right)^3 + \dots + \left(z^{21} + \frac{1}{z^{21}}\right)^3$ is _____.

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$21 + \left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \left(z^3 + \frac{1}{z^3}\right)^3 + \dots + \left(z^{21} + \frac{1}{z^{21}}\right)^3$ is _____.

13.00

Solution:

$$z = \frac{1-i\sqrt{3}}{2} = e^{-\frac{\pi}{3}}$$

$$z^r + \frac{1}{z^r} = 2 \cos\left(-\frac{\pi}{3}\right)r = 2 \cos \frac{r\pi}{3}$$

$$\Rightarrow 21 + \sum_{r=1}^{21} \left(z^r + \frac{1}{z^r}\right)^3 = 8 \left(\cos^3 \frac{r\pi}{3}\right) = 2 \left(\cos r\pi + 3 \cos \frac{r\pi}{3}\right)$$

$$\Rightarrow 21 + \left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \dots + \left(z^{21} + \frac{1}{z^{21}}\right)^3$$

$$= 21 + \sum_{r=1}^{21} \left(z^r + \frac{1}{z^r}\right)^3$$

$$= 21 + \sum_{r=1}^{21} \left(2 \cos r\pi + 6 \cos \frac{r\pi}{3}\right)$$

$$= 21 - 2 - 6$$

$$= 13$$

Q90: The number of three-digit even numbers, formed by the digits 0, 1, 3, 4, 6, 7 if the repetition of digits is not allowed, is _____.

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52.00

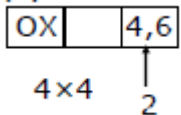
Solution:

(i) When '0' is at unit place



Number of numbers = 20

(ii) When 4 or 6 are at unit place



Number of numbers = 32

So number of numbers = 52

